Does User Engagement Shape Congressional Candidates' Issue Agendas on Twitter?

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Abstract

Does user engagement shape the issue agendas of congressional candidates on Twitter? We analyze over 3.4 million tweets from all candidates in the 2020 and 2022 election cycles to estimate the engagement returns to discussing different political topics and test whether those returns influence which issues candidates emphasize. We find that tweets referencing divisive issues — such as guns, abortion, race, Trump's impeachment, January 6th, Biden, and China — receive substantially greater engagement than tweets about the issues voters care most about, including the economy, health care, education, climate, and Covid-19. Yet these incentives do not shape candidates' agendas. In aggregate and at the individual level, candidates predominantly discuss voter-relevant issues, and even those who receive especially large engagement boosts from divisive topics are no more likely to discuss them. We provide evidence that diminishing engagement returns to discussing divisive issues partly explain this restraint and theorize that representation and accountability further tether candidates' agendas to voters' priorities. Engagement amplifies polarizing issues online, but it does not determine the distribution of issues that candidates choose to emphasize.

Keywords: *Social Media*, *Campaigns*, *Engagement*

Word Count: 8,850

1 Introduction

Understanding how politicians participate in social media is important for two reasons. First, politicians play a direct role in shaping public discourse. Many have wide followings, and their posts can set the day's political agenda for the media and broader public, influencing which topics get discussed and in what tone. Second, their behavior on social media provides a window into how they see the world and what they prioritize.

One force widely believed to shape that behavior is user engagement: candidates may seek likes and retweets for electoral gain, fundraising, or the pleasure of being the center of attention. Much of what we know about these incentives comes from research on tone. A vast literature demonstrates that users engage more with posts that are negative (Hansen et al. 2011), incivil (Heseltine and Dorsey 2022), polarized (Finkel et al. 2020; Brady et al. 2017), or feature out-group animus (Ballard et al. 2022). This scholarship implies that candidates will have incentives to supply more negative and conflictual content, not only for direct increases in engagement, but also for the downstream engagement returns from social media algorithms boosting high-engagement posts (Guess et al. 2023).

Whether politicians have engagement incentives to discuss certain *topics* over others is important as well. Topic choice can carry downstream effects: for example, if politicians emphasize crime and immigration, voters may come to evaluate politics in those terms (Iyengar et al. 1982; Iyengar and Kinder 1987, chapter 7). Yet such topic incentives have received much less attention in the US context, and an even bigger gap lies in whether politicians' issue agendas actually respond to them (see Section 2).

In this paper, we explore the demand and supply sides of the topics US politicians discuss on social media. On the demand side, we assess which topics generate the most engagement from users. On the supply side, we assess which topics candidates emphasize the most, exploring whether their agendas cater to social media demand or to voters' priorities as expressed in public opinion surveys.

To do so, we examine over 2.2 million tweets from 1,207 congressional candidates in the 2020 election cycle and over 1.25 million tweets from 1,275 candidates in 2022. We inductively identify the language associated with candidates' high- and low-engagement tweets, define thirteen political topics from these terms, and estimate each topic's association with engagement using fixed-effects count models. Related models yield candidate- and time-specific effects, which, combined with survey data from YouGov and Pew Research Center, allow us to test whether candidates' issue agendas better reflect their engagement incentives or their voters' priorities.

We find that although users disproportionately engage with tweets related to divisive

national issues, such as guns, abortion, and January 6th, congressional candidates still mostly emphasize issues that matter most to voters, such as the economy, health care, and education. Even the candidates who receive the greatest boosts in engagement from divisive issues are no more likely to discuss them. This holds even among challengers, a group that must gain attention to electorally compete. Ideologically extreme and electorally safe candidates are more likely to prioritize divisive issues, yet they have no greater engagement incentives to do so.

Why do candidates leave engagement on the table? We provide evidence that diminishing returns partly explains this behavior. We show that the engagement return to discussing a divisive issue typically declines as a candidate repeatedly discusses it, so an agenda reoriented toward divisive content could not sustain the returns that made it attractive. However, diminishing returns are only part of the story: even when heavily used, divisive topics still generally outperform the issues voters prioritize. We theorize that representation and accountability supply the remaining constraint — candidates tether their issue agendas to voters' priorities in order to be perceived as authentic, in step, and electorally viable.

Our results suggest that the role of engagement in candidates' issue agendas is nuanced. By itself, it is concerning that polarizing topics are boosted, as members of the political public, beyond just Twitter users, will see this content before tweets about more productive topics. However, this engagement incentive does not meaninfully shape the distribution of topics candidates choose to emphasize. This goes against the implication in the tone literature, and a common refrain in the broader public, that political candidates produce social media content primarily to grab attention. Online campaign discourse is skewed towards divisive issues because of audience demand more so than candidate supply.

2 Virality, Strategic Communication, and Issue Agendas

We are interested in understanding the topics that politicians produce for social media for multiple reasons. First, as Bene (2021) points out, both the agenda-setting (Iyengar et al. 1982) and priming (Iyengar and Kinder 1987, chapter 7) theories of mass media argue that exposure to some topics over others affects citizens' cognitive political processes. A classic example is H.W. Bush's negative advertising, which effectively forced the 1988 presidential election to be a referendum on Dukakis' supposed soft stance on crime. Second, understanding the topics politicians emphasize helps us gain insight into their priorities and goals – for example, which audience they prioritize when communicating their issue agendas.

We focus on two sides of political topic production. On the demand side, we focus on

potentially competing audience demands: what users engage with and what voters prioritize. On the supply side, we want to understand which audience politicians' issue agendas primarily respond to.

2.1 Demand Side: User Engagement

While many have found that negative and conflictual tone gets higher levels of engagement (Hansen et al. 2011; Heseltine and Dorsey 2022; Finkel et al. 2020; Brady et al. 2017; Ballard et al. 2022), subject matter may also affect social media engagement. In the American context, this relationship has received far less attention than tone, though researchers have documented suggestive term-level patterns.

Pew researchers used data from 2015-2020 to identify 100 terms for each party that increase engagement on both Congressional tweets and Facebook posts by at least ten percent (Van Kessel et al. 2020). Republicans boosted engagement by using terms related to immigration and China, while Democrats boosted engagement by using terms related to health care and gun safety. Shah and Grant (2021) take a more limited approach, identifying the top 10 terms that received outsized engagement for each party in 2016 and 2020. They find that Democrats did best when discussing Trump and Supreme Court Justice Barrett, while Republicans did best when mentioning election integrity, Castro, and Hunter Biden. Because these studies examine a small number of individual terms rather than entire topics, however, it is difficult to use their results to determine which topics candidates have the greatest engagement incentive to emphasize, or how issues rank against one another.

Some research has sought to embed the discussion of topic effects into political frameworks. For example, Chang et al. (2025) tested whether liberal and conservative users engage differently with rhetorical "toxicity" and topics "owned" by the out-party. They report that "Democrat-dominated topics" performed best for members of Congress (MCs) in both parties (p. 12), where a party owns a topic if its MCs sent more tweets discussing it. Democratic-owned topics in their classification include the environment, civil rights, education, health, crime, immigration, social welfare, domestic labor, technology, housing, and transportation. Because this approach pools many issues into a single ownership bin, it cannot identify which topics actually drive engagement — indeed, we find that many of these "Democrat-owned" topics perform poorly for members of both parties.

The relationship between political topics and engagement has been studied more directly in other national contexts, most often on Facebook (Heidenreich et al. 2022; Bene et al. 2022; Fenoll et al. 2024). For example, Bene (2021) analyzed political topic choice in the Hungarian context and found that on Facebook, posts related to corruption and campaigning receive

outsized engagement, and Blassnig et al. (2021) analyzed the Facebook pages of political parties in Germany and Austria in 2017 and found that parties' top issues increased shares and that identity issues increased comments.

We cannot assume such findings port directly to the American Twitter context, as engagement dynamics differ across both platforms and national contexts (Larsson 2026; Stier et al. 2018). For instance, in a study of parties' Facebook posts during national election campaigns in Germany, Hungary, and Norway, Tønnesen et al. (2025) find that posts addressing the economy and the environment are among those driving user reactions — quite unlike the patterns we document for American congressional candidates on Twitter.

Regardless, our contribution lies less in documenting which topics engage than in comparing users' demands with those of politicians' other key audience, voters, as well as testing whether engagement returns ultimately inform candidates' issue agendas.

2.2 Supply Side: Candidate Topic Priorities

Understanding what kind of content generates engagement is not sufficient to understand social media activity by political candidates. Candidates have multiple goals. Typically, we think of their primary goal as maximizing electoral chances. High levels of engagement may support electoral gains by boosting name recognition, para-social ties to supporters and perhaps increasing fundraising (Ballard et al. 2022). But candidates may also wish to appeal to voters' concerns, which may diverge from the issues that appeal to social media users.

Interviews with campaign staffers provide mixed evidence about how candidates choose their social media content. Kreiss et al. (2018) interviewed digital staffers from seven 2016 presidential campaigns and found that all campaigns paid data analytics services to closely monitor which issues and tones performed best and "adjusted communicative practices accordingly" (p. 17). At the same time, several of these same campaigns explicitly said they do not chase engagement. For example, Bernie Sanders' strategist said, "it was never about how witty we could be to get something to go viral...It was always about let's just talk about the issues that Bernie would be talking about" (p. 14). Echoing the same sentiment, Ted Cruz's strategist affirmed "The most important things for us rather than something performing well [in terms of audience metrics] was just that it stayed within our message and within Senator Cruz's voice... every time a post went viral it was 'by accident' " (p. 14) (see also Klinger and Russmann (2017)).

Because it is hard to generalize from well-funded presidential campaigns to congressional campaigns, research that empirically assesses how candidates choose their content is valuable. Several papers examine the distribution of topics political officeholders discuss on Twitter

(e.g., Barbera et al. 2019; Shapiro and Hemphill 2017). In addition, Barbera et al. (2019) find that members of the 113th Congress increased discussion of certain issues when their followers and/or the media began discussing them. But responding to current events does not imply candidates update based on what generates engagement. More importantly, temporarily shifting attention toward an issue when it becomes topical is conceptually distinct from structuring one's *overall* issue agenda around the pursuit of attention. These are related but separate estimands, and we return to this distinction in the discussion.

To our knowledge, Frimer et al. (2023), Schöll et al. (2024), and Ennser-Jedenastik et al. (2022) are the only studies that empirically test whether campaigns strategically respond to engagement incentives. They generally find evidence that politicians adjust their content in the short-run based on engagement. Thus, like Barbera et al. (2019), these studies provide evidence of marginal, dynamic updating — again, an important but distinct estimand. They also examine very different contexts: incivility rather than issue agendas (Frimer et al. 2023), an Austrian Facebook campaign observed for six weeks and roughly 1,500 policy-relevant posts (Ennser-Jedenastik et al. 2022), and Spanish MPs on gender issues specifically Schöll et al. (2024). It is unclear whether these dynamics extend to American politicians and the full distribution of issues they emphasize across a campaign.

In sum, questions remain about how politicians choose the topics they post online about. Do they follow user demand? Or does their behavior track with voters' concerns? Do these patterns vary according to context?

3 Which Issues do Voters Prioritize?

To orient theoretical expectations on these questions, we must first establish which issues voters prioritize. We therefore analyze survey data from YouGov, which tracks issue priorities among a representative sample of registered voters in every week of our study period (YouGov 2023). We supplement it with Pew surveys: although these provide only snapshots, they allow respondents to select multiple prioritized issues and ask about a couple issue areas YouGov does not (Covid-19 and welfare programs), providing additional evidence and a robustness check (Pew Research Center 2019, 2020, 2021, 2022c,a).

While partisan differences certainly exist, a consistent categorization yet emerges. Figure 1 shows that both parties' voters generally prioritize a similar set of issues — the economy, health care, and education, along with climate for Democrats specifically — while a second set — abortion, gun control, race, and crime — generally ranks lower.

For Democratic voters, health care, the economy, and climate change are far and away the

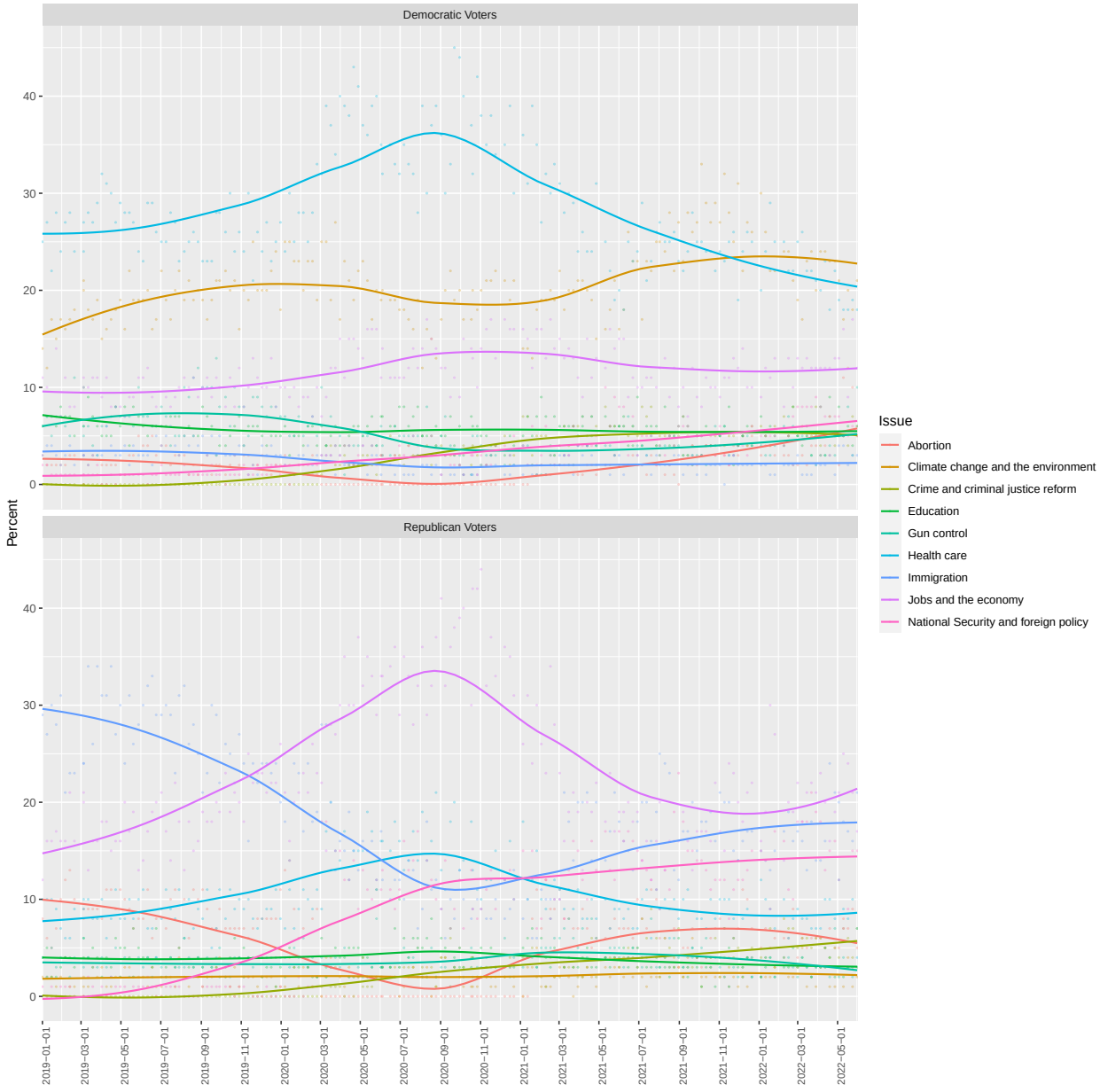


Figure 1: Share of Voters by Party Identifying Issue as "Most Important". *Source*: YouGov Weekly Tracker. Each weekly survey asks a representative sample of registered voters from YouGov's US Research Panel to identify the single issue most important to them. Respondents are weighted according to age, gender, race, education, and 2016 Presidential vote. The demographic weights correspond to the proportion identified in the most recent American Community Survey. The total weighted sample of Democratic and Republican respondents in each weekly survey ranges from 373 to 904 (median = 399) and from 235 to 611 (median = 306), respectively. Curves are LOESS smoothers.

most important issues; even education polls as equally or more important than abortion, gun control, crime and immigration. The Pew snapshots (Figure 2) replicate these patterns and further show that Democrats rank Social Security and Covid-19 above crime, immigration, race, abortion, and guns, with minimal exceptions. Indeed, even in Pew's October 2022 survey — after abortion and gun policy became hyper-salient with the *Dobbs* decision and the school shooting in Uvalde, Texas — Democrats still placed health care above abortion, and education, climate, and the economy above gun policy.[1]

Compared to Democrats, Republicans place greater emphasis on immigration, less on climate, and increasingly identify foreign policy as a top concern after the coronavirus pandemic began. Still, the economy is almost always their first (only briefly their second) most important issue, and across both election cycles health care polls well above abortion, gun control, and crime, while education generally polls equal to or above them as well. Pew largely corroborates this picture: Republicans consistently rank the economy most important, and they label health care, education, and Social Security as more important than race and often similarly important to immigration, crime, abortion, and guns. Before 2022, Republicans also ranked Covid-19 above race and, at times, above immigration and crime as well.

3.1 Categorizing Issues: Material and Divisive

The two sets that emerge from these surveys fit a substantive distinction rooted in existing theories of issue conflict. We refer to the relatively prioritized issues — the economy, health care, education, Covid-19, and climate — as *material* issues, for they appear more directly tied to voters' everyday needs. We refer to the less-prioritized set — issues like guns, abortion, and race — as *divisive* issues. On material issues, voters broadly agree on the ends (for example: more jobs, better schools, less Covid-19, a cleaner environment) and primarily debate the means, but on divisive issues, the ends themselves are contested. One can simply be pro- or anti-abortion, gun control, or immigration (Carmines and Stimson 1980), and the debate more often features battles over core values (Johnston and Wronski 2015). Thus, while debates on material issues often feature more technical stances on the means (for example, the optimal approach to create jobs), voters can more easily take stances on divisive issues by leveraging their emotional, cultural, and value-system priors (Carmines and Stimson

[1] Regardless, the jump in issue saliency does not concern our categorization of voters' priorities, because our data ends May 31, 2022 and the *Dobbs* decision was in June 2022 and Uvalde shooting was May 24, 2022.

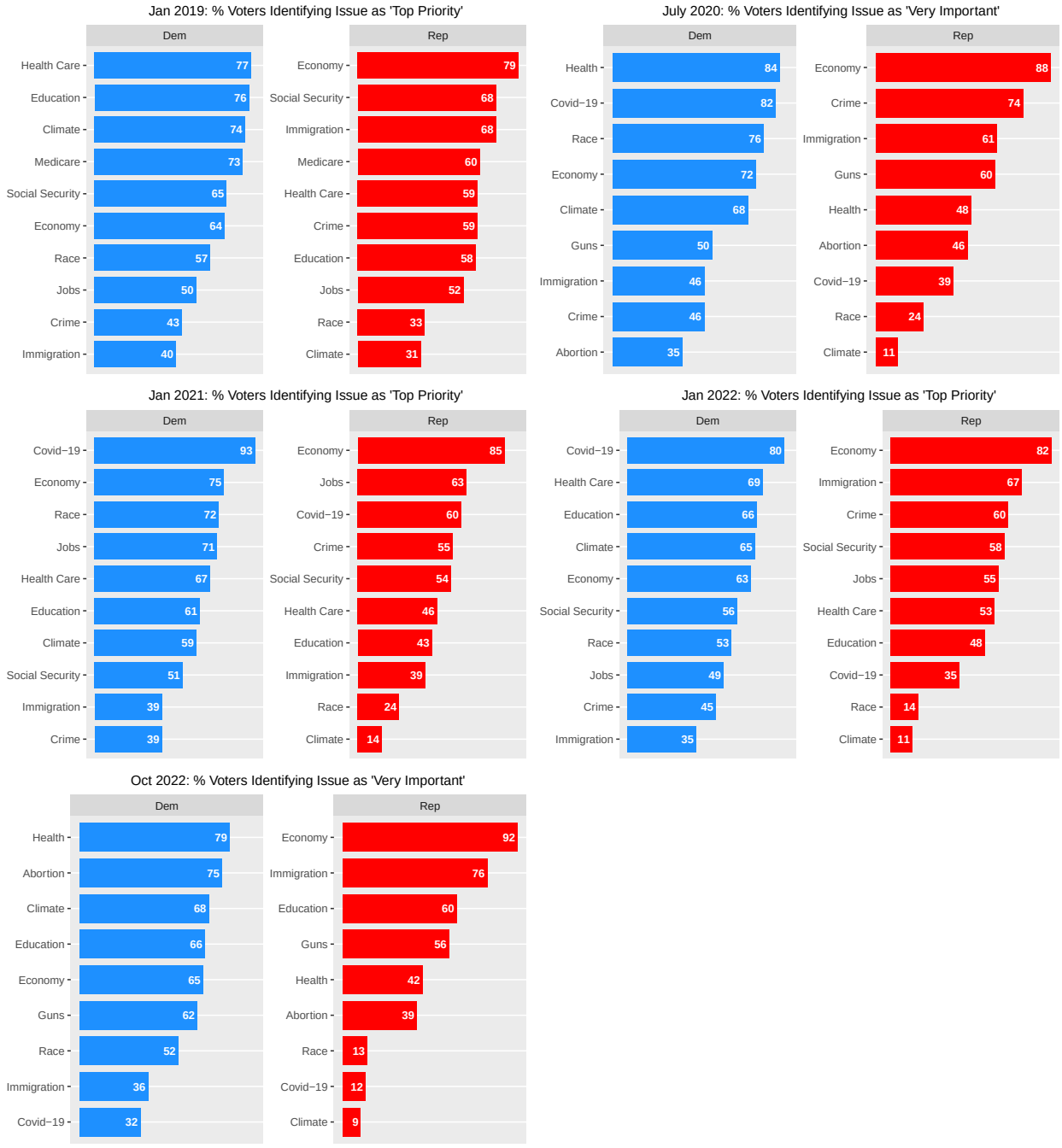


Figure 2: Share of Voters by Party Identifying Issue as "Very Important" or as a "Top Priority". Respondents may select multiple issues: surveys ask which issues are a "top priority" to them for the contemporary presidential administration and Congress, or are "very important to their vote". Sample sizes of each survey in chronological order: 1,505, 11,001, 5,360, 5,128, 3,993. *Source*: Pew Research Center.

1980).[2] These issues thus lend themselves to being more polarizing, symbolic, emotional, and associated with extreme attitudes (Fenoll et al. 2024).

Such issue categorizations are common in the literature (Simons and Green 2018; Elliott-Dorans 2022; Kim et al. 2018). Fenoll et al. (2024, p. 4), for example, separate "consensual" issues (the analogue of our material set), such as education, health care, and infrastructure, from divisive issues such as gun control, race, and abortion.

Of course, material issues can be divisive at times, and vice versa. But crucially, the takeaways from our analyses will not hinge on these qualitative labels: readers who dislike them may simply substitute "voter-prioritized" and "less-prioritized" for "material" and "divisive" issues, respectively, throughout.

4 Theoretical Expectations

Having reviewed the literature in Section 2 and established voters' priorities in Section 3, we now state our expectations, first for user demand and then for candidate supply.

Given the many findings that Twitter users engage most with divisive tone, some evidence from other national contexts that divisive topics generate greater engagement (see Section 2), and the fact that Twitter users differ demographically from voters (McClain et al. 2021; Pew Research Center 2022b), we expect users' demands (as expressed through engagement) and voters' demands (as expressed in surveys of their issue priorities) to diverge:

> **Hypothesis 1** (Competing Audience Demands)**.** *Users are more likely to engage with divisive issues than with the issues voters prioritize.*

We expect this divergence to vary across candidates, as politicians may differ in their communication strategies for electoral, organizational, and personality reasons (Nai and Maier 2024; Ennser-Jedenastik et al. 2022). Ideological extremists, seeking to develop and maintain their ideological brand (Clarke 2020), and electorally secure candidates, less reliant on courting their districts' marginal voters, may both have incentives to cultivate national and ideologically charged audiences. We therefore theorize that their follower networks will reward divisive content most:

> **H1.1:** *The divergence between user engagement and voter priorities is largest for ideological extremists and candidates in electorally safe seats.*

[2] Carmines and Stimson (1980) use the labels "easy" versus "hard". Based on their definitions, our "divisive" and "material" issues fall cleanly under easy and hard, respectively.

If users indeed reward divisive content, which audience do candidates' issue agendas serve? We state two competing supply-side hypotheses. The first is consistent with the literature on audience capture in social media (DiResta 2024), and with the reviewed literature implying, sometimes without direct empirical testing (e.g., Heseltine and Dorsey 2022; Ballard et al. 2022), that politicians will supply more of the content that generates attention. Seeking electoral viability, fundraising, and/or name recognition, candidates have an incentive to tilt their agendas toward the issues that generate engagement.

> **Hypothesis 2** (User-Driven Supply)**.** *(a) Candidates devote a greater share of their tweets to divisive issues than to the issues voters prioritize. (b) In particular, candidates whose divisive-issue tweets are associated with greater engagement returns allocate a greater share of their agenda to these issues.*

Challengers have the greatest electoral and financial need for attention, as they typically lack name recognition, institutional support, and established donor networks. This perhaps makes engagement particularly valuable to them. Thus, we expect that if any candidates chase engagement, it should be challengers:

H2.1: *Challengers' issue agendas are the most responsive to engagement returns.*

Alternatively, candidates' agendas may track the voters they seek to represent rather than the users who reward them:

> **Hypothesis 3** (Voter-Driven Supply)**.** *(a) Candidates devote a greater share of their tweets to the issues voters prioritize than to divisive issues. (b) Candidates with larger engagement returns to divisive issues are no more likely to discuss them.*

Why might candidates leave engagement on the table? We theorize two complementary mechanisms.

The first relates to *representation* and *accountability*. All else equal, campaigns likely prefer more engagement to less, but interview evidence from Kreiss et al. (2018) suggests they may avoid chasing engagement for fear of appearing inauthentic. Particularly if campaigns also struggle to discern which issues systematically perform best (Kreiss et al. 2018), then the representational returns to discussing voter-relevant issues is the safer investment.

The second is *diminishing returns*: if engagement boosts are driven by novelty and event-driven salience, the engagement returns to divisive issues cannot be scaled — a candidate who reoriented their agenda toward divisive content would watch its returns erode. This mechanism yields a direct prediction, which we test in Section 6.2.3: a topic's engagement returns should decline with greater recent usage.

5 Empirical Approach

We have two goals: estimate the engagement return to discussing a topic and then evaluate whether these incentives shape candidates' issue agendas. This section details how we select topics to analyze and quantify their association with engagement. Heterogeneity across candidates in engagement returns, as well as analyses of the downstream consequences of these incentives — such as how they relate to candidates' issue agendas and voter priorities — build directly on this framework, so we introduce and discuss them in their respective results sections.

5.1 Data

We use the Python package `twint` and Twitter API to scrape tweets for all congressional incumbents and challengers from the 2020 (used `twint` for this data) and 2022 (used API for this data) election cycles, including both campaign and official handles for incumbents. The 2020 data spans from January 2019 to December 2020; the 2022 data spans from January 2021 to May 2022.[3][4] We limit our analysis to candidates who received at least two percent in their primary election and/or were general election candidates. We also only examine those candidates with at least 50 tweets. For the 2020 election cycle, we analyzed 1,207 candidates and over 2.2 million tweets. Among these candidates, 232 are Democratic incumbents, 413 are Democratic challengers, 188 are Republican incumbents and 367 are Republican challengers. For the 2022 election cycle, we have 1,275 candidates and over 1.25 million tweets. Among which are 250 Democratic incumbents, 387 Democratic challengers, 217 Republican incumbents and 409 Republican challengers.

We measure engagement as the sum of retweets and likes per tweet, excluding replies because they are sometimes used to criticize rather than amplify content (Shugars et al. 2021).

[3] Our data does not include retweets for both practical and theoretical reasons. Practically, it is difficult to calculate engagement on retweets because the Twitter API does not provide "like" counts for them. Theoretically, including retweets may be problematic because they are not originally authored content and may not reflect content the candidate endorses. Many officials note in their Twitter bios that they do not necessarily endorse the content of their retweets (Hemphill et al. 2013).

[4] Tweets were scraped after allowing at least three days to pass, at which point, significant changes in engagement are rare.

5.2 Identifying Topics by Identifying Engagement Terms

Because political topics are not directly observable in text, specifying which issues to classify ex ante risks imposing an arbitrary structure on the analysis. We therefore induct terms associated with candidates' high- and low-engagement tweets, and from these terms, construct topics to classify. Our topics are informed by engagement patterns, not researcher priors, because we are not interested in topics with neutral engagement effects. Rather, we seek to identify issue areas that plausibly generate engagement incentives, since our goal is to examine whether such incentives shape issue agendas.

We use the tf-idf algorithm to identify high- and low-engagement terms. Compared to other unsupervised approaches, such as topic models, tf-idf is desirable because it is designed to identify distinctive terms. It is given by

$$w_{i,d} = tf_{i,d} * log(N/df_i), \tag{1}$$

where term frequency ($tf_{i,d}$) is the number of times term i appears in document d as a proportion of all terms in d and the inverse document frequency is the inverse of the ratio defined by the number documents term i appears in (df_i) divided by the total number of documents (N). To be distinctive, a term must appear frequently in a given document and infrequently in others.

We first define high-engagement tweets as those receiving a 100% or more increase in engagement relative to a candidate's mean, and low-engagement tweets as those receiving a 75% or more decrease in engagement relative to a candidate's mean. For each candidate, we then gather all high engagement tweets into one document and all low engagement tweets into another. We focus on within-candidate comparisons, because candidates vary substantially in the number of followers they have and in their overall political status. After removing stopwords, non-English words, and merging terms into relevant n-grams[5], we apply tf-idf and collect the top 50 distinctive terms (highest tf-idf scores) from each document. This process yields a list of high- and low-engagement terms for each candidate. We then group candidates by party, incumbency status, and election cycle, and within each group, identify the 25 most commonly occurring terms in candidates' individual lists of high- and low-engagement terms.

A potential concern with applying tf-idf to only two documents is that any term appearing at least once in both receives a score of zero, even if it is much more prevalent in one. As a result, tf-idf may omit terms that are typically, but not exclusively, associated with high or low engagement. However, by restricting attention to terms that are unique to a candidate's

[5]For example, the four words "make america great again" become one term "make_america_great_again".

Table 1: Top 25 High and Low Engagement Terms for 2020 Incumbents, alphabetized

Dem High Engagement	Dem Low Engagement	Rep High Engagement	Rep Low Engagement
116thcongress	access	abortion	access
abused	assistance	adam_schiff	assistance
barr	benefits	alive	care
condemn	businesses	amy_coney_barrett	communities
constitution	census	articles_of_impeachment	community
donald_trump	county	babies	coronavirus
endtheshutdown	deadline	born	county
hr8	economic	ccp	critical
impeach	funding	censure	economic
impeachment	health	communist	families
inquiry	info	consent	family
joebiden	information	discharge	health
kamalaharris	infrastructure	endinfanticide	information
mueller	issues	gopleader	issues
oath	local	hong_kong	local
presidentelect	program	impeachment	news
presidents	programs	inquiry	program
racist	register	letusvote	resources
resign	resources	petition	service
shutdown	service	reelection	services
sotu	services	schiff	students
sworn	small_businesses	sham	testing
trial	students	stevescalise	update
usps	town_hall	stoptheschiffshow	veterans
wall	veterans	unanimous	women

high- or low-engagement document, we can be more confident in the associations identified within each candidate. This is important because we later aggregate these candidate-level lists: if individual lists contain weak or ambiguous associations, the aggregation step risks amplifying noise rather than signal. By prioritizing accuracy over coverage, our approach aligns with the goal of this inductive step—identifying the language most clearly associated with engagement outcomes.

For incumbents, clear patterns emerge across parties and election cycles. High-engagement terms generally refer to provocative, polarizing national issues, while low-engagement terms relate to issues prioritized by voters in surveys. Table 1 shows that for Democrats in 2020, discussing the highly partisan issues of Trump's impeachment, Biden, guns and background checks[6] proved effective. However, discussing education, small businesses, veterans, coron-

[6]"hr8" is the Bipartisan Background Checks Bill.

Table 2: Top 25 High and Low Engagement Terms Among 2022 Incumbents, alphabetized

Dem High Engagement	Dem Low Engagement	Rep High Engagement	Rep Low Engagement
cabinet	assistance	abortion	access
capitol	communities	capitol	act
constitution	community	ccp	assistance
convict	county	communist	businesses
donald_trump	critical	constitution	communities
election	eligible	constitutional	community
equalityact	federal	democracy	county
impeach	funding	election	critical
impeachment	health	elections	economy
inauguration	information	electors	families
incited	local	gopleader	family
inciting	pandemic	hamas	funding
insurrection	payments	impeach	industry
joebiden	program	impeachment	information
mob	programs	iran	infrastructure
oath	project	israel	issues
officer	receive	keystone	legislation
overturn	register	relief	local
police	resources	stevescalise	military
president_trump	service	taiwan	news
remove	services	unacceptable	program
removed	students	unity	resources
trump	town_hall	violence	service
violent	vaccine	votes	students
voted	veterans	voting	veterans

avirus testing, and health and economic services – all issues that tended to be prioritized by Democratic voters – led to lower engagement. Terms related to accessing public services, such as "program", "service", "access", "funding", and "resources", also had low traction. Even directly referring to local constituents (e.g., "communities", "county", "town hall") lowered engagement.

Table 2 shows that the pattern was similar for Democrats in 2022. Divisive national topics, like the events of January 6th, Trump's second impeachment, equity legislation, and the filibuster, were associated with higher engagement. Terms related to education, health services, vaccination, workers, and so on were associated with lower engagement.

Tables 1 and Table 2 show similar patterns for Republican incumbents in both election cycles. Controversial national terms, like impeachment (both Trump's first and second impeachment), abortion, communism, China, January 6th, Hamas, and Israel garnered more engagement. Partisan framing against the 2019 impeachment and abortion, labeling it a 'sham' and 'infanticide', respectively, was also successful.[7] The low engagement terms are remarkably similar to the Democrats, including terms related to students, veterans, inflation, businesses, health care, COVID-19, services, and the economy.

Tables 3 and 4 suggest that, for challengers, although similar patterns remain, it is less clear what differentiates high- and low-engagement terms. Challengers from both parties gain from certain controversial terms (e.g., "black lives matter" and "gun" for Democrats or "abortion" and "socialism" for Republicans) and lose engagement when discussing issues like Covid-19, the economy, and services. But many of the high engagement terms are relatively neutral, and in the 2022 election cycle results, they sometimes even refer to "material" issues prioritized by voters. Terms related to health, education, and Covid-19 ("healthcare", "teachers", and "school" for Democrats; "school" and "vaccine" for Republicans) are associated with high engagement.

In addition, some terms appear in both the high- and low-engagement lists. In Table 4, for example, terms related to family and gender for Democrats ("families", "children", "women") and politically-charged terms for Republicans ("biden", "left", "republicans") are associated with both high and low engagement. Because we calculate the most-occurring terms across all candidates' tf-idf lists, this means that for some significant group of challengers in each party, these terms work, but for another group, they fail. Challengers exhibit less consistency than incumbents in the kinds of terms that receive high and low engagement.

Across parties and election cycles, these results suggest that incumbent congressional can-

[7]Calling the 2019 impeachment a "sham" was a conservative rallying call which deemed the entire impeachment inquiry a political ploy (Hains 2019). And "endinfanticide" is a conservative slogan supporting the Born Alive Act that implies Democratic politicians do not oppose doctors killing infants who survive abortions (Ollstein 2019).

Table 3: Top 25 High and Low Engagement Terms Among 2020 Challengers, alphabetized

Dem High Engagement	Dem Low Engagement	Rep High Engagement	Rep Low Engagement
ballot	access	american	american
blacklivesmatter	communities	americans	americans
constitution	community	ballot	bill
democrat	county	conservative	care
democratic	covid19	constitution	community
donald_trump	crisis	elected	county
elect	donate	election	covid19
election	education	endorsement	democrats
election_day	facebook	freedom	economy
endorsement	family	god	election
equality	food	life	families
grassroots	free	pelosi	family
gun	health	police	issues
joebiden	information	polls	job
kamalaharris	issues	president	leadership
nominee	leadership	prolife	life
polls	local	radical	local
president	love	republican	love
primary	policy	rights	money
spent	public	socialism	news
trump	realdonaldtrump	trump	party
usps	resources	values	republican
voters	students	voters	service
voting	women	voting	veterans
washington	workers	washington	world

Table 4: Top 25 High and Low Engagement Terms Among 2022 Challengers, alphabetized

Dem High Engagement	Dem Low Engagement	Rep High Engagement	Rep Low Engagement
abortion	american	americans	american
ballot	americans	biden	americans
ban	care	bill	biden
bill	children	border	children
child	communities	children	county
county	community	conservative	democrats
democrat	county	county	families
democratic	covid	endorsement	family
election	democracy	families	freedom
endorsement	donate	family	god
families	families	freedom	gop
gun	family	left	inflation
healthcare	issue	policies	job
justice	issues	president	leadership
leader	life	republican	left
president	local	republicans	life
republican	love	rights	love
rights	money	school	money
school	party	schools	party
spent	pay	ukraine	policies
teachers	power	vaccine	power
unseat	service	values	president
voters	vote	voters	republican
voting	women	voting	vote
women	workers	washington	world

didates experience greater engagement when they use terms related to controversial national issues rather than those directly relevant to voters. But it is unclear to what extent these patterns exist for challengers. Challengers exhibit greater heterogeneity in the language that sells, as divisiveness does not always predict term-engagement associations.

5.3 Classifying Topics

Based on these tf-idf results, we identify 13 engagement-relevant topics: Trump, Biden, Covid-19, race, climate, education, health, economy, guns, abortion, China, 2019 impeachment, and January 6th. The impeachment topic refers to Trump's first impeachment in 2019. Thus, for the 2020 data, we include impeachment as a topic, not January 6th, and vice versa for the 2022 data.

We use a dictionary-based method to code tweets into these non-mutually exclusive categories. For example, tweets about race include "Black Lives Matter", "George Floyd", "All Lives Matter", "minorities", and so on, while tweets about Covid-19 include "covid19", "coronavirus", "pandemic" etc. Tweets can also be a combination of topics if they include keywords of different topics. See Table A.1 for our dictionaries' list of terms.

While we could use LLMs to classify these tweets, we use dictionaries because they are computationally inexpensive and well suited to our use case. Political topics require much less contextual understanding to detect than latent dimensions like sentiment and polarity (which dictionaries perform poorly on), so we can reliably map keywords to a given topic. For example, tweets containing the terms "background checks" or "second amendment" almost certainly concern the topic "guns". Other topic classification studies have achieved high accuracy with such methods (Dun et al. 2021; Soroka and Wlezien 2019; Baker et al. 2016; Stephens-Davidowitz 2014; Muddiman et al. 2018). See Dun et al. (2021) for further discussion of the merits of dictionaries when topics contain "readily-identifiable" words.

We follow best practices from the classification literature to ensure our dictionaries perform well. We avoid the ill-advised, but common practice of using "off-the-shelf" dictionaries from other research. This approach is liable to incorrect classification, as keywords in one substantive domain may not align with how words are used in another (Grimmer and Stewart 2013). We therefore follow Muddiman et al. (2018) and curate our own dictionaries specifically designed to identify topic-related terms in congressional tweets. Such curation is not a trivial task, as references to our topics do not just occur in buzzwords but also through slang and hashtags. By tailoring our dictionary to the specific language environment of political Twitter, we ensure the domain specificity that Grimmer and Stewart (2013) call for.

Specifically, we initialized a set of logical keywords for each topic and then sampled tweets

to iteratively improve the dictionaries. We then validated our final dictionaries on a separate hand-coded sample of 2000 tweets (1000 randomly sampled from each election cycle). Table 5 provides validation metrics. As with most dictionaries, we have high precision and accuracy but lower recall. However, our F1 scores, with a mean of 0.80, compare favorably against other studies using supervised ML methods to classify tweets into a large number of categories (e.g. Payson et al. 2022). Accuracies approach 1 because, relative to the number of tweets, most topics are mentioned relatively infrequently.

Table 5: Validation Metrics

Topic	Accuracy	Precision	Recall	f1
Climate	0.98	0.88	0.62	0.73
Biden	0.99	0.94	0.85	0.89
Trump	0.99	0.84	0.92	0.88
Race	0.98	0.84	0.77	0.80
Education	0.98	0.77	0.77	0.77
Covid-19	0.99	0.92	0.89	0.91
Economy	0.94	0.75	0.78	0.76
Health	0.97	0.72	0.72	0.72
Guns	1.00	0.92	0.80	0.86
China	1.00	0.92	0.92	0.92
Abortion	1.00	0.94	0.89	0.91
Impeachment_2020	0.99	0.86	0.75	0.80
January 6th	0.99	0.78	0.70	0.74

[1] Metrics are rounded to the second decimal point

5.4 Quantifying Topic-Engagement Associations

We use negative-binomial regressions of the following specification to estimate the relationship between topics and engagement. For tweet w,

$$Y_w = \beta_0 + \sum_{j=1}^{12} \beta_j T_w^{(j)} + \boldsymbol{\eta}\mathbf{X} + \alpha_c + \delta_t + \varepsilon_w \quad (2)$$

where Y_w is total engagement (likes + retweets) on tweet w, $T_w^{(j)}$ is a dummy variable in-

dicating whether tweet w mentions topic j, and α_c and δ_t are candidate and time fixed effects, respectively. $\mathbf{X}$ is a vector of controls for incivility, defined as swear words and ad hominems, and polarizing partisan phrases, language frequently associated with the Trumpian right and progressive left which we call Trumpism and Leftism, respectively, and include terms like "maga", "america first", or "defund the police" (see Appendix A for full set of included terms). These covariates not only help control for polarizing tone, but also tell us whether simply mentioning token partisan phrases is associated with engagement even after controlling for a tweet's political substance.

We estimate Equation 2 with negative-binomial (NB) regression, because engagement is a count outcome with a sizable share of zero-counts and overdispersion — the variance of engagement exceeds its mean by several orders of magnitude — rendering OLS inappropriate. NB regression directly models the count structure while flexibly accommodating this overdispersion, making it a standard estimator in this setting (Wooldridge 2010). Our findings do not hinge on this choice: estimating the models by OLS with a log(1 + engagement) outcome produces substantively similar results (Figures A.13 and A.14).

Our findings also do not hinge on modeling candidates with fixed rather than random effects: Figures A.15 and A.16 show the point estimates are identical with random effects.[8] We prefer fixed effects here from a theoretical perspective, because our data is the population of congressional candidates' tweets in these cycles. We are therefore much more interested in an unbiased estimate of topic-engagement associations within our sample, than in incurring in-sample bias for a model that can predict out-of-sample. For unbiasedness, random-effect models require candidate effects to be uncorrelated with topic choice, a dubious assumption given that candidates' baseline engagement levels likely correlates with their issue portfolios.

Finally, our findings are not an artifact of combining likes and retweets into a single engagement measure. Re-estimating the main models with likes and retweets as separate outcomes yields nearly identical topic effects (Appendix Figures A.17–A.20): across all 120 topic coefficients, the main-text estimates correlate at 0.99 with the likes-only estimates and 0.98 with the retweets-only estimates.

We estimate eight separate models split by party, challengers and incumbents, and election cycle. Recall that for models estimated in the 2020 election cycle, we include `Impeachment` instead of `January 6th` as a covariate, and vice-versa for the 2022 election cycle. Standard errors are clustered at the candidate level.

This model is descriptive but quite controlled, as we make within-candidate comparisons

[8]The random- and fixed-effect estimators converge when the within-group sample sizes grow large. This is the case in our data: the median candidate in the 2020 and 2022 election cycles sent 778 and 594 tweets, respectively.

while accounting for common time trends. In appendix models, we also account for changes in follower counts, the primary within-candidate time-varying confounder, and the results are nearly identical. We also make minimal parametric assumptions, as the model simply estimates differences-in-means. Recall that a tweet can be labeled as multiple topics, so there is no held-out topic category. One should interpret $(e^{\beta_j} - 1) * 100$ as providing the within-candidate and within-month percentage change in engagement associated with discussing topic j, holding other topic and tone categories fixed.

We will apply variants of this specification in later sections to get candidate-level coefficients for heterogeneity analyses, as well as month-level coefficients for comparison against voters' priorities and candidates' issue agendas.

6 Results

6.1 Issues Relevant to Voters Get Less Engagement

Two key points are important to consider when interpreting the regression results. First, statistical significance is less informative than the magnitude of the estimates, as even small shifts in engagement may be statistically significant given the large sample size. Second, considering we are classifying charged political topics, we should expect the coefficients to vary but remain positive. Many candidate tweets, particularly from official Twitter handles, are mundane and procedural. Thus, a significant negative coefficient, signaling a decrease in engagement associated with a topic is particularly intriguing.

Recall that if candidates are incentivized to discuss issues prioritized by their party's voters, then for Democratic candidates, tweets related to climate, the economy, health, education, and Covid-19, should all increase engagement. Climate, economy, and health-related content in particular should garner more attention than issues like gun policy, abortion, race, and China (to the extent that Democratic voters' interest in "foreign policy" expressed in surveys reflects interest in China, see Figure 1). For Republican candidates, tweets related to the economy should receive the most attention. Health- and education-related tweets should garner similar levels of engagement to issues like gun control and abortion, and greater engagement than race.

Figure 3 suggests this is not the case, as we find support for H1: polarizing national issues significantly outperform the issue areas prioritized by voters (see Figure A.11 for model coefficients on the percent-change scale). For incumbents of both parties across both

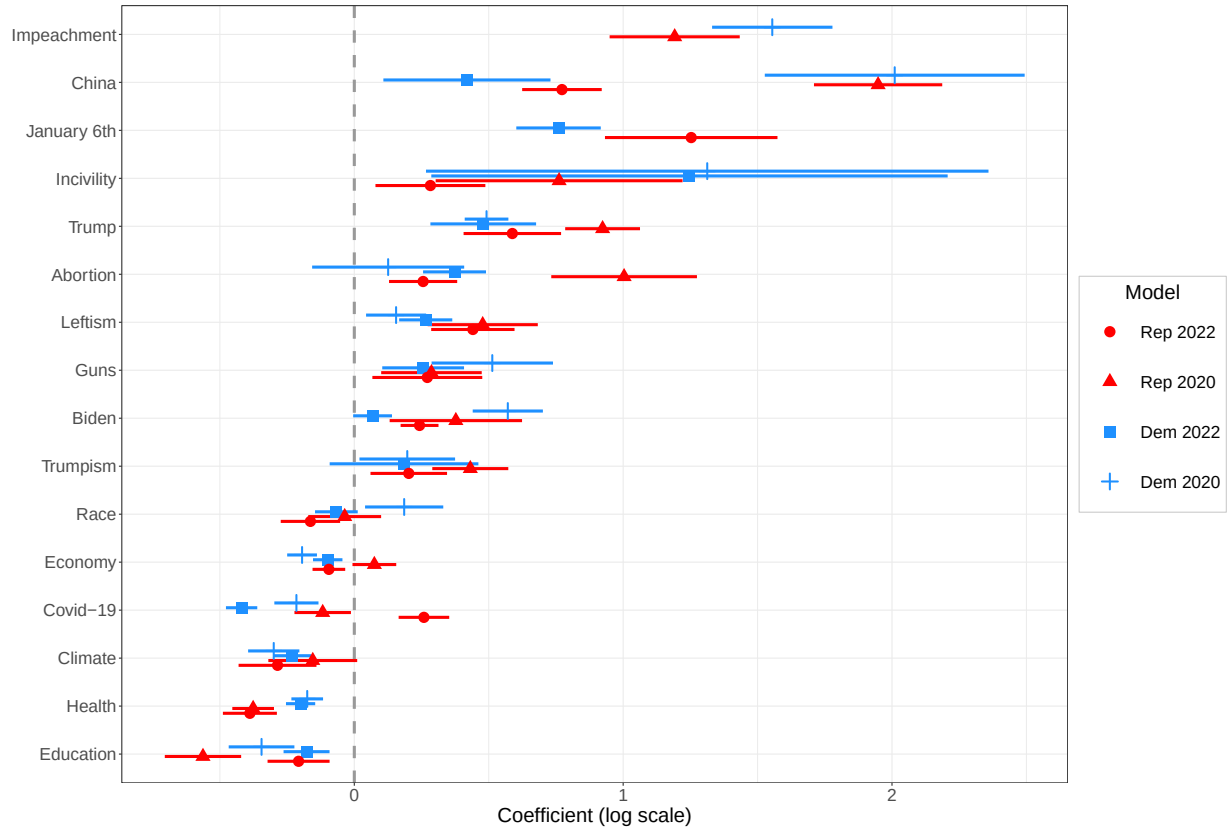


Figure 3: Association between Topics and Engagement for Incumbents. Point estimates are the regression coefficients from four separate models split by party and election cycle. Topics are ordered by their average coefficient value across the four models. We cluster standard errors at the candidate-level. $(e^{\beta_j} - 1) \times 100$ gives the within-candidate, within-month percent change in engagement associated with discussing topic j, holding other topic and tone categories fixed. See Figure A.11 for the coefficients on the percent-change scale, and Table A.2 for the regression table.

election cycles, Trump, Biden, guns, abortion, impeachment, January 6th, and China are typically associated with significant increases in engagement. For example, tweets referencing Trump, Biden, China, abortion, and guns received from 27% to 646% greater engagement, with most coefficients falling between roughly 30% and 120%.[9] Similarly, averaging across parties, discussing the 2019 impeachment and January 6th is associated with a 300% and 180% increase in engagement, respectively.

These patterns do not apply to issues voters prioritize. Across both parties and election cycles, climate, education, health, the economy, and Covid-19 are associated with engagement losses or statistically insignificant returns — the lone exception being Covid-19 among Republicans in 2022, which is associated with a 30% engagement increase. For example, averaging across election cycles, discussing health or education is associated with a 31% engagement loss for Republican incumbents and a 20% loss for Democratic incumbents; Democratic tweets referencing climate or the economy suffer similar losses, while Republican tweets on the economy deliver near-zero returns.

Figure 4 shows that different engagement dynamics exist for challengers. Unlike for incumbents, coefficients on some of the issues prioritized by voters are now positive and significant. For example, among Democrats, discussing education, the economy, or health is associated with a 36%, 33%, and 17% within-candidate engagement increase, respectively, averaged across election cycles; and among Republicans in the 2022 election cycle, tweets related to health, education, and Covid-19 received 28%, 35%, and 52% increases, respectively. These findings fit with our original hypothesis that all politically relevant speech should increase engagement, given that most tweets by Congressional candidates are procedural and mundane.

Although all else equal, discussing certain voter-prioritized issues is associated with engagement increases for challengers, tweets related to more polarizing topics still perform best. January 6th, impeachment, guns, and race are still the most viral topics. For example, averaging across election cycles, Democratic challengers' tweets on guns and race saw 54% and 43% engagement increases, respectively; for Republican challengers, the corresponding increases were both 42%. Recall that voters of both parties significantly prioritize the economy over these issues, while Democrats also care much more about health and climate. Challengers may increase their online visibility by tweeting about certain issues most relevant to voters' concerns, but like incumbents, they are still incentivized to discuss divisive social issues when they most want to go viral.

[9] Only abortion among Democratic incumbents in 2020 and Biden among Democratic incumbents in 2022 are statistically insignificant at the 5% level, though their point estimates remain positive.

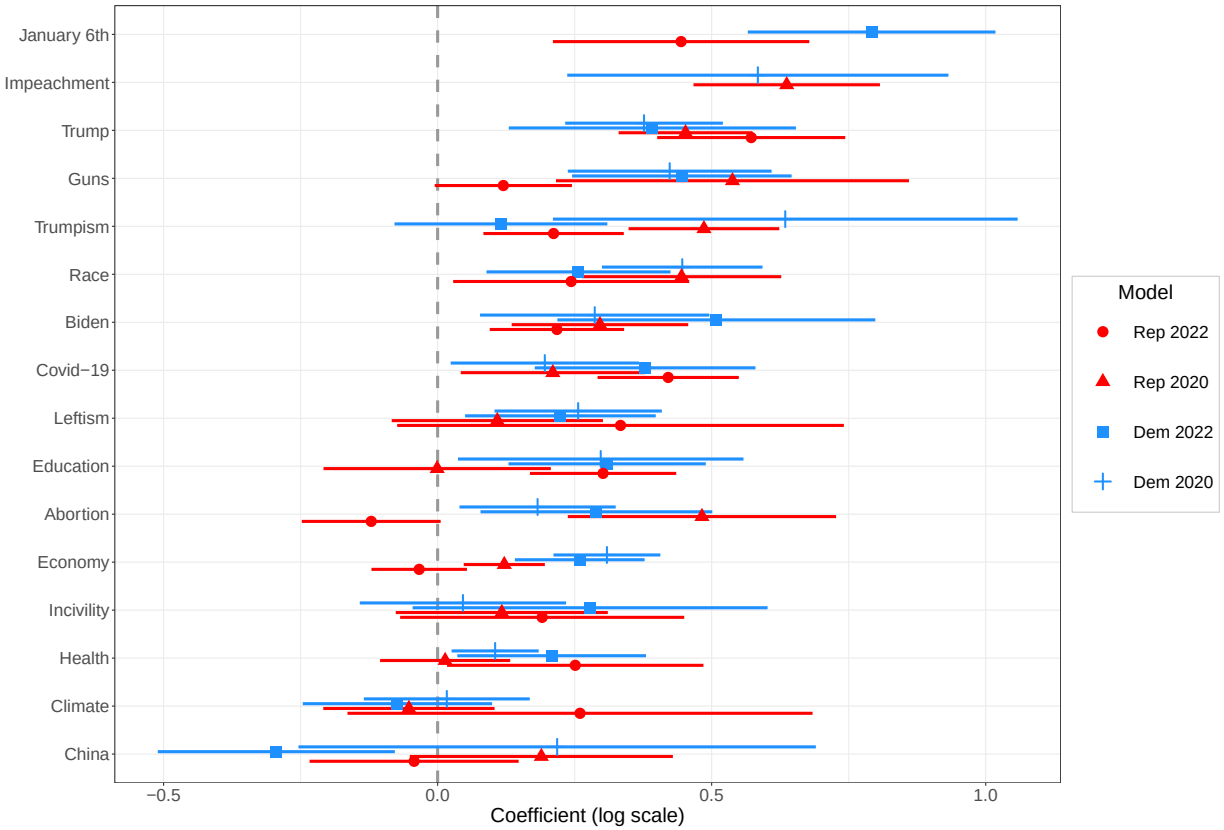


Figure 4: Association between Topics and Engagement for Challengers. Point estimates are the regression coefficients from four separate models split by party and election cycle. Topics are ordered by their average coefficient value across the four models. We cluster standard errors at the candidate-level. $(e^{\beta_j} - 1) \times 100$ gives the within-candidate, within-month percent change in engagement associated with discussing topic j, holding other topic and tone categories fixed. See Figure A.12 for the coefficients on the percent-change scale, and Table A.3 for the regression table.

6.1.1 Engagement Incentives Constant Across Time and Candidate Types

These results are largely consistent across different candidate types, and we thus do not find support for H1.1. In Appendix C, we show that incumbents facing different levels of electoral vulnerability see similar returns across the full set of issues, as do challengers across the ideological spectrum. Among incumbents, a few divisive topics — China, impeachment, guns (Democrats only), and January 6th (Republicans only) — show ideological moderates reaping the largest gains; on ideology, then, the evidence if anything cuts *against* H1.1. But on most issues, particularly those voters prioritize most, returns are comparable across the ideological spectrum.

In Appendix E, we provide a more granular comparison of voters' priorities and user demand, as both may shift over an election cycle. By leveraging the weekly YouGov survey data and month-level engagement models, we compare voters' and users' issue rankings *within each month*. We find that the divergence holds in each month of our data: voters consistently prioritize the economy, health care, and education (and for Democrats, climate as well), while users spurn these issues in favor of issues less immediately concerning to voters in those months, such as abortion and guns. Hence, across the election cycle, candidates face competing demand incentives from these two audiences.

6.2 Do These Engagement Incentives Shape Issue Agendas?

The tone literature often implies that politicians are encouraged to employ polarizing rhetoric because such content gets greater engagement. However, the extent to which politicians adjust their online profiles towards the content receiving greatest engagement remains unclear. If this mechanism is correct, we expect congressional candidates to prioritize divisive issues because they receive greater attention.

6.2.1 Aggregate Comparison

Figure 5 plots the average share of a candidate's tweets devoted to each topic and shows that, in aggregate, this is not the case. We find evidence supporting H3(a) and opposing H2(a): candidates primarily discuss material issues, even though they receive relatively little attention. Among incumbents, across parties and election cycles, the economy, health, and Covid-19 are generally the most discussed topics, while impeachment, China, January 6th, abortion, and guns — issues associated with the largest engagement returns — are minimally discussed. Among Democrats, mentions of education and climate also outpace these high-

engagement topics. Similar patterns exist for challengers, except Republicans place greater weight on race than health.[10] The conclusions are identical when examining the total volume of tweets on each topic (Appendix Figure A.1).

6.2.2 Candidate-Specific Comparisons

For a more granular analysis, we now compare engagement incentives to topic prioritization at the individual candidate level. We group topics into material or divisive and, within each party × incumbency × cycle subsample, estimate a negative-binomial model in which the candidate fixed effects are interacted with the two issue-type indicators:

$$Y_w = \beta_0 + \sum_c D_{c,w} \left(\alpha_c + \beta_{1,c} \text{Material}_w + \beta_{2,c} \text{Divisive}_w \right) + \mathbf{X}\boldsymbol{\eta} + \delta_t + \epsilon_w \tag{3}$$

where $D_{c,w}$ indicates whether tweet w was sent by candidate c. Each candidate thus has their own intercept — the candidate fixed effect α_c — and their own issue-type slopes $\beta_{1,c}$ and $\beta_{2,c}$, which measure candidate c's within-candidate engagement returns to material and divisive tweets, net of the common month effects δ_t.[11] $Material_w$ indicates whether a tweet mentions Covid-19, climate, education, health, or the economy, while $Divisive_w$ indicates whether a tweet mentions guns, abortion, the 2019 impeachment (for the 2020 cycle only), January 6th (for the 2022 cycle only), race, China, Trump, or Biden. Because we allow tweets to mention multiple topics, these binned categories are not mutually exclusive. As in our main specification (Equation 2), $\mathbf{X}$ controls for incivility and polarizing partisan phrases.

The material/divisive categorization is simply one way to group these issues; neither our core argument nor the subsequent results rely on these qualitative labels. The material issues closely track those that voters consistently report are most important to them (with the exceptions that Republican voters do not prioritize climate and have waxed and waned on Covid-19), while the divisive issues are less prioritized. If one does not like these qualitative labels, one can therefore simply view the two groups as voter-prioritized and less-prioritized issues. We use the material-divisive categorization because the issues voters do and do not prioritize generally reflect this substantive distinction (see Section 3.1).

[10] The only divisive topic that gets discussed significantly is mentions of Trump in 2020 (among both parties) and Biden in 2022 (among Republicans). However, mentions of sitting presidents and presidential candidates function less as policy areas than topics like the economy or health care, and more as focal points of partisan conflict. Given that tweet volume on other issue areas appears unrelated to their engagement return, outsized mentions of Biden and Trump likely reflect the nationalization of congressional elections and the dominance of presidential politics, rather than a strategic shift toward a policy issue.

[11] We fix the dispersion parameter at the subsample-level estimate from our main models; the candidate-specific coefficients are insensitive to this choice.

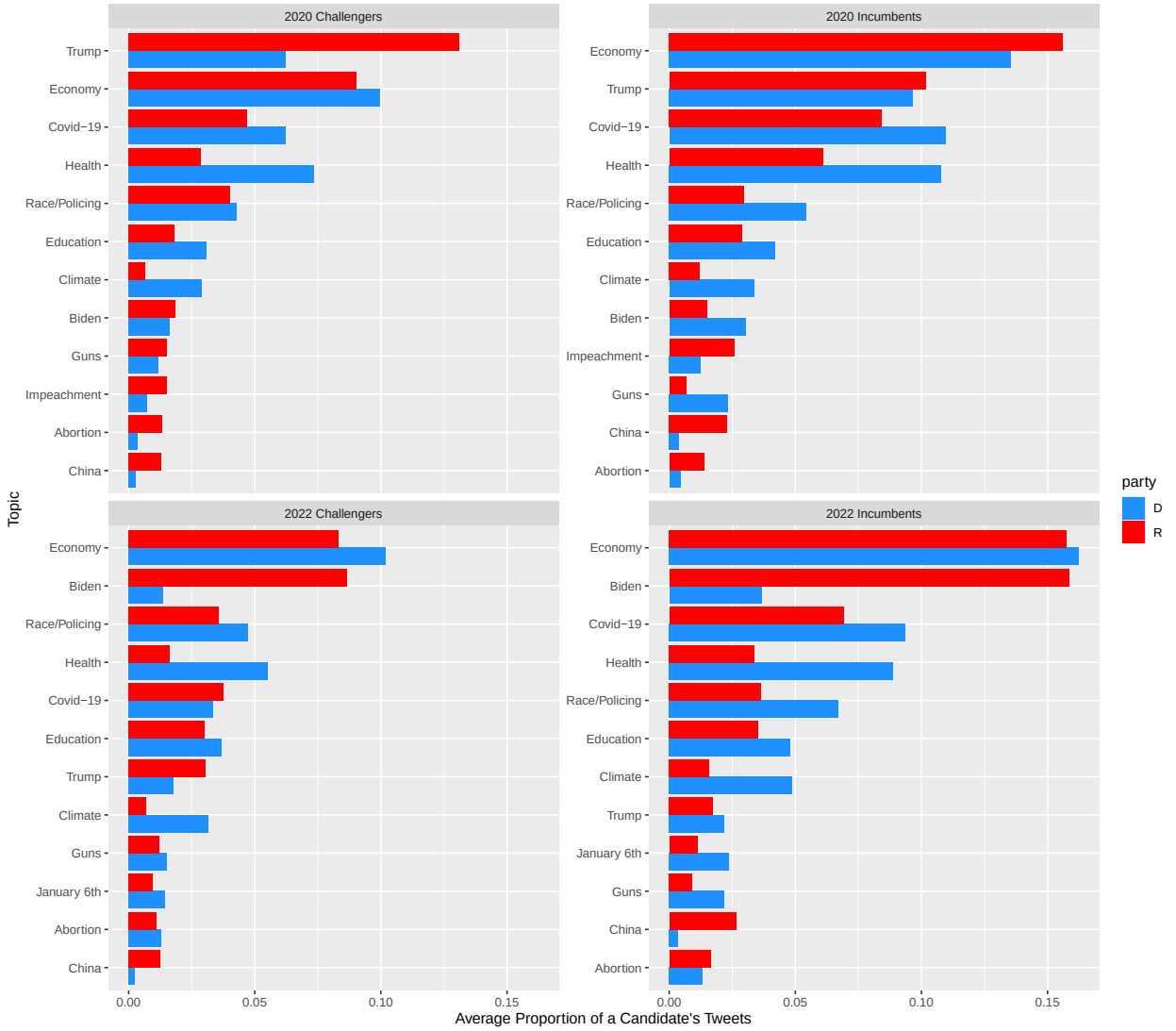


Figure 5: Candidates Primarily Discuss Material Issues. Bars show the average share of a candidate's total tweets mentioning each topic, averaged across candidates within each party × incumbency × cycle stratum. Within each pane, topics are ordered top-down by the average of the two parties' bars; the tone categories (Trumpism, Leftism, and incivility) are omitted. In our data, the 2020 election cycle spans January 2019–December 2020, and the 2022 election cycle spans January 2021–May 2022.

Figure 6 plots the relationship between candidate c's estimated $\beta_{1,c}$ ($\beta_{2,c}$) and the proportion of c's total tweets composed of material (divisive) issues — split by party, incumbency status, and election cycle. Let $y_{1,c}$, $y_{2,c}$ and $\hat{\beta}_{1,c}, \hat{\beta}_{2,c}$ denote candidate c's proportions and estimated coefficients. If candidates' issue agendas are shaped by which topics perform best, then *at least* one of the following must hold: (1) the correlation between $\hat{\beta}_{1,c}$ and $y_{1,c}$, or $\hat{\beta}_{2,c}$ and $y_{2,c}$ is positive, and/or (2) $\overline{\hat{\beta}}_i > \overline{\hat{\beta}}_j$ and $\bar{y}_i > \bar{y}_j$ (where the averages are taken over all candidates c within a party-incumbency-cycle stratum).

In words, the first condition says candidates with larger engagement returns to discussing divisive (larger $\beta_{2,c}$) or material (larger $\beta_{1,c}$) topics, spend a greater proportion of their tweets on divisive or material issues, respectively. Visually, this implies the green or orange curve slopes upwards. The second condition says the average candidate receives greater engagement from one issue type and sends a greater proportion of their tweets on that issue type. Visually, this implies one issue-type's point cloud is in the upper right corner, while the other is in the lower left. Either a within- or between-issue type relationship between engagement and topics supplied is necessary.

Neither of these conditions hold; the evidence is consistent with H3(b) rather than H2(b) and H2.1. First, across parties, incumbency status, and election cycles, candidates with particularly large engagement incentives to discuss divisive or material issues appear no more likely to do so. Second, it is not the case that one issue type relative to the other is associated with larger engagement returns *and* a greater share of total tweets. For example, among incumbents, although divisive issue coefficients are clearly larger than material issue coefficients (the average orange dot is to the right of the average green dot), they do not make up a larger proportion of candidates' tweets. In fact, if anything, other than Republican challengers, material issues are discussed *more* (evidenced by the smooth fits, which show the conditional mean of total tweet proportion for material issues is generally higher than that of divisive issues across the support of x).

Then who, if anyone, discusses divisive issues in particular? Figures A.5-A.7 show that among incumbents, progressives and conservatives, as well as those in safe districts, focus a greater share of tweets on divisive issues. Crucially, this is *not* driven by engagement returns, as these legislators have no greater engagement incentive to discuss these issues — if anything, it is ideologically *moderate* incumbents who see the largest returns on several divisive topics (see Figures A.2–A.4). Among challengers, topic agendas are not only uniform across engagement incentives but also across the ideological spectrum, which is perhaps consistent with their greater idiosyncrasy due to greater independence from the party infrastructure.

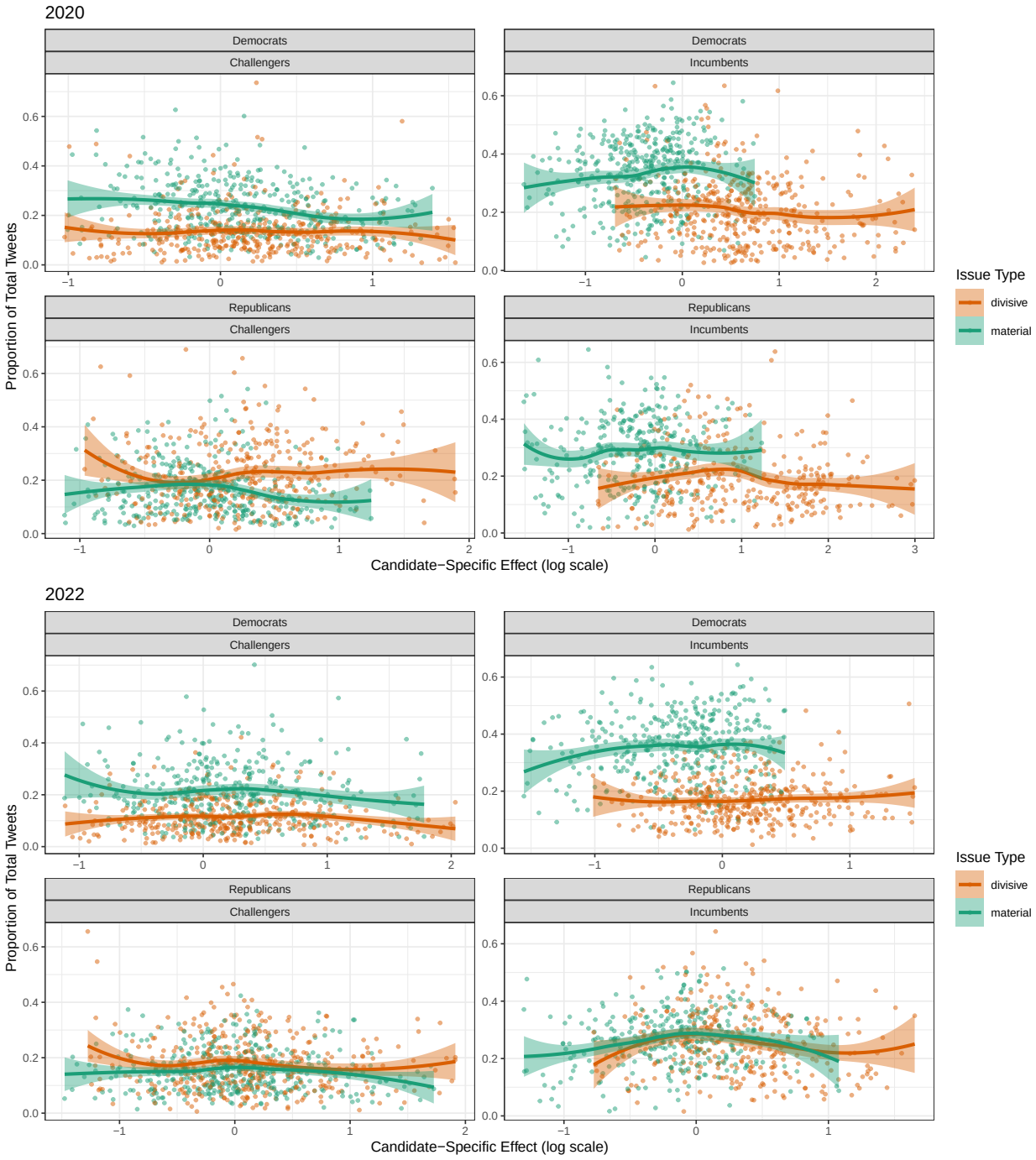


Figure 6: Candidates with Larger Engagement Incentives to Discuss Certain Topics are Not More Likely to do so. Within issue-type, each dot is a candidate's estimated issue-type coefficient (x-axis, log scale) against the share of their total tweets spent on those issues (y-axis). To limit the influence of a few extreme outliers, each panel displays the central 95% of the coefficient distribution (2.5–97.5 percentiles, computed within party-incumbency-cycle-issue-type cells). Figure titles denote the election cycle. Curves fit using LOESS.

6.2.3 Mechanisms: Diminishing Returns to Repetition

If divisive content reliably generates engagement, why do candidates not reallocate their agendas toward it? One possibility, theorized in Section 4, is that the returns documented in Figures 3 and 4 are not available at scale: if the engagement a topic generates depends partly on its novelty, then repeatedly discussing it should erode its returns, leaving candidates little to gain from restructuring their agendas around viral topics. If novelty contributes to engagement in this way, then a topic's engagement boost should decline with how intensively a candidate has recently discussed it. We test this directly. For tweet w by candidate c, let $S_{j,w}$ denote the share of the candidate's 50 preceding tweets that mention topic j (the current tweet excluded). Within each party × candidate-type × cycle subsample, we estimate

$$Y_w = \beta_0 + \sum_{j=1}^{12} \beta_j T_w^{(j)} + \sum_{j=1}^{12} \gamma_j \left(T_w^{(j)} \times S_{j,w} \right) + \boldsymbol{\eta}\mathbf{X} + \alpha_c + \delta_t + \varepsilon_w \tag{4}$$

where Y_w is engagement on tweet w.[12] The coefficient β_j on topic j's main effect gives its engagement association when the candidate has not recently discussed the topic, and $\gamma_j < 0$ indicates diminishing returns to repetition. We drop each candidate's first 50 tweets so that S is always defined over a full window. Figures 7 and 8 plot the topic-engagement associations as a function of S, $\hat{\beta}_j + \hat{\gamma}_j S$, for the 2020 cycle; the 2022 results for both candidate types are similar (Appendix F).

Among incumbents (Figure 7), the topics commanding the largest engagement boosts show stark diminishing returns. For example, fresh impeachment tweets — those sent by incumbents who had not recently discussed the topic — are associated with a roughly 450–500% engagement increase, but the return declines steeply with recent intensity and is statistically indistinguishable from zero at the highest observed intensities; Trump and Biden tweets follow the same pattern. These are event-driven topics: they pop when novel and fall off when candidates double down. By contrast, guns and abortion — long-standing divisive policy areas rather than event-driven controversies — if anything show *rising* returns, though the narrower range of observed intensities on these topics cautions against strong conclusions. Notably, at virtually any observed intensity, the divisive issues (even the more event-driven ones) continue to outperform the voter-prioritized issues — China, for example, still returns roughly 400% increases for incumbents who devoted 40% of their recent tweets to it. This suggests the outsize attention divisive substance receives is driven by more than just novelty and event-driven salience — though we do not directly show it here, their divisiveness itself is plausibly another component. The voter-prioritized issues instead display mildly rising

[12] The lower-order terms $\sum_{j=1}^{12} \lambda_j S_{j,w}$ are included in the model but suppressed in Equation 4 for visual clarity.

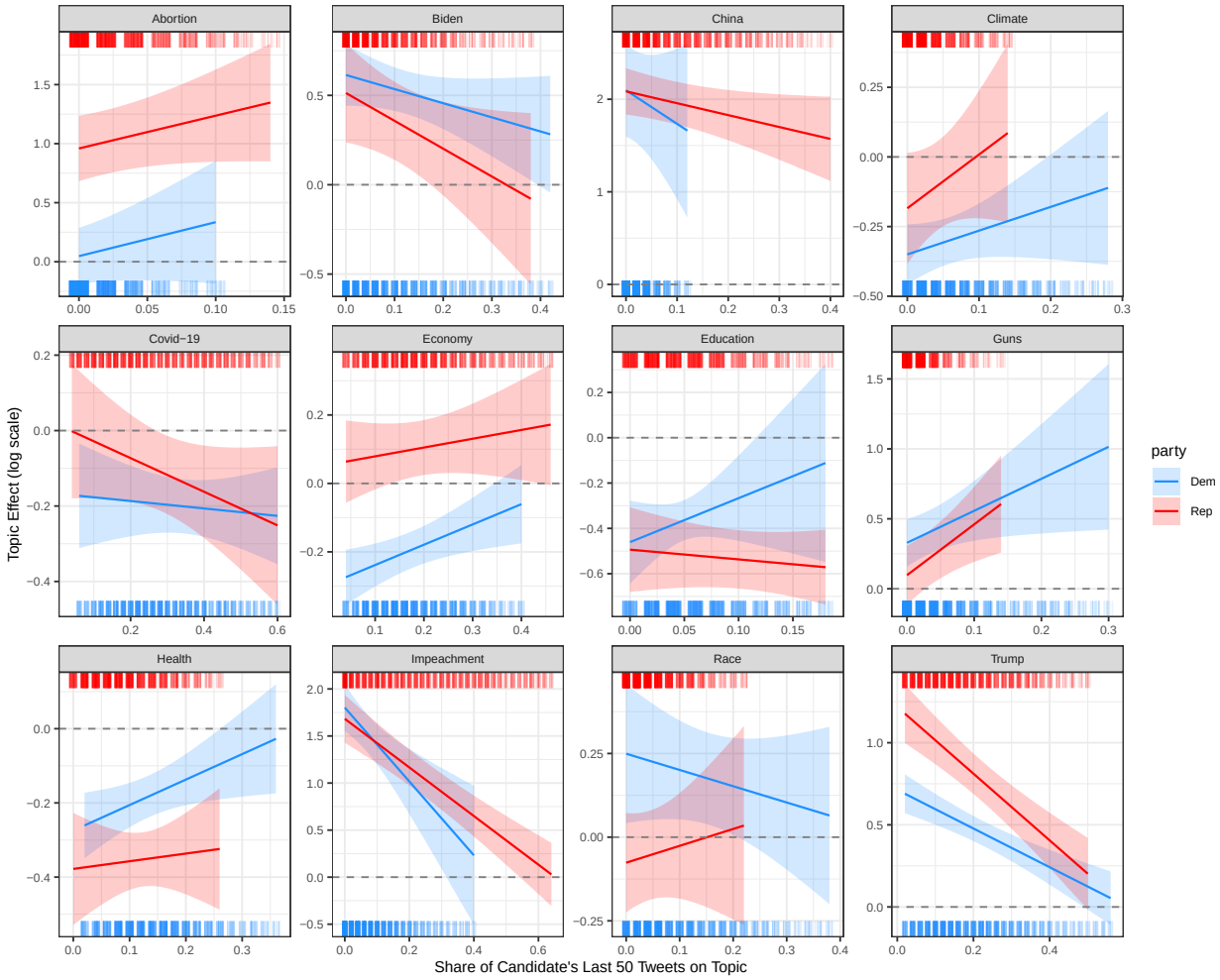


Figure 7: Diminishing Returns to Repetition Among 2020 Incumbents. Lines plot the implied effect of each topic ($\hat{\beta}_j + \hat{\gamma}_j S$, Equation 4) as a function of the share of the candidate's 50 preceding tweets mentioning that topic, over the central 90% of the observed intensity distribution among the topic's tweets. Bands are 95% confidence intervals from candidate-clustered standard errors. Rug marks show the observed intensity distributions (Democrats on the bottom edge, Republicans on the top edge).

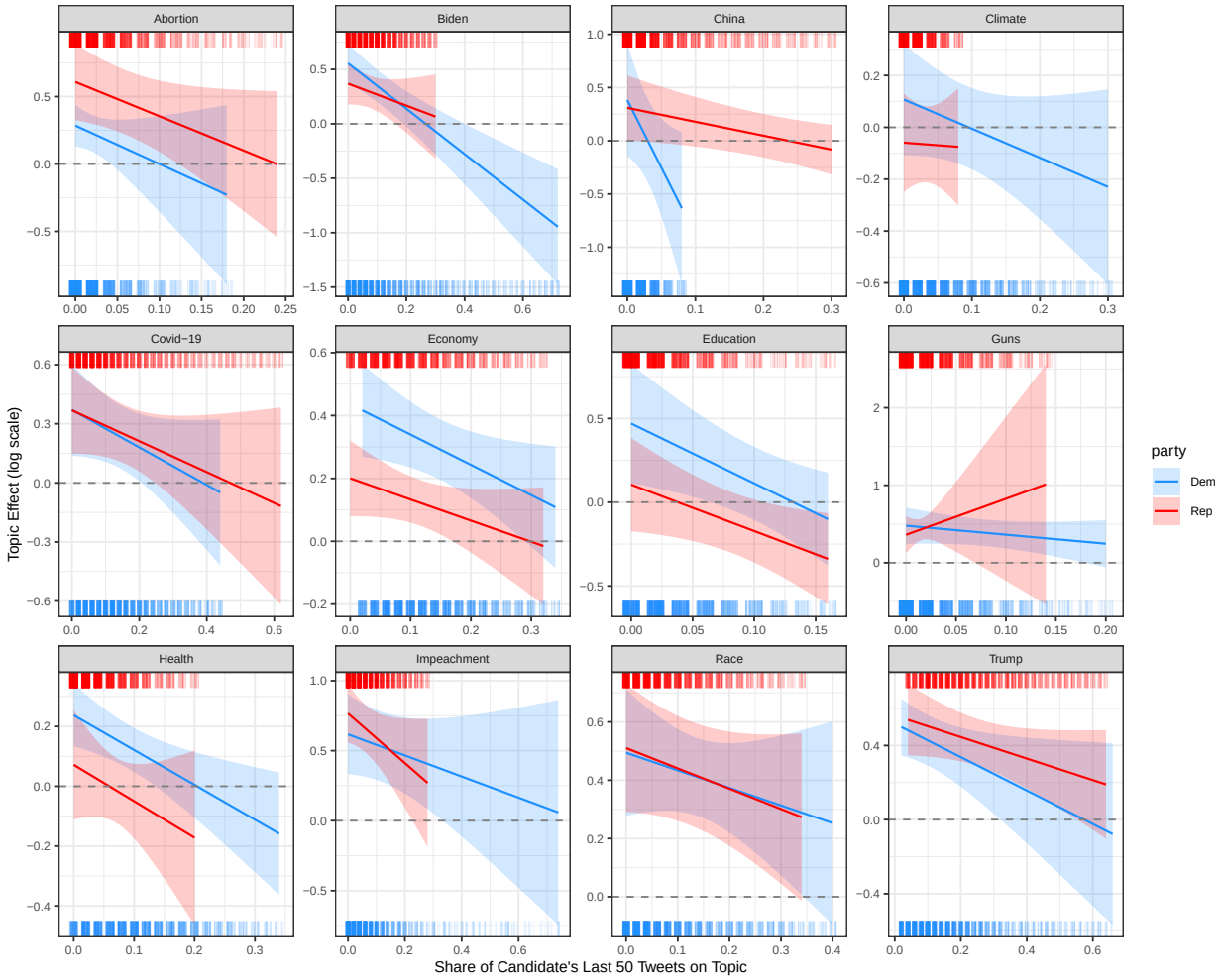


Figure 8: Diminishing Returns to Repetition Among 2020 Challengers. Lines plot the implied effect of each topic ($\hat{\beta}_j + \hat{\gamma}_j S$, Equation 4) as a function of the share of the candidate's 50 preceding tweets mentioning that topic, over the central 90% of the observed intensity distribution among the topic's tweets. Bands are 95% confidence intervals from candidate-clustered standard errors. Rug marks show the observed intensity distributions (Democrats on the bottom edge, Republicans on the top edge).

profiles that at best approach a null effect at high intensity, potentially because candidates discuss them most intensively precisely when they are contemporaneously salient, partially offsetting users' general disinterest.

Among challengers (Figure 8), returns diminish nearly across the board, including on the voter-prioritized issues. This is consistent with challengers have more insular networks: the same audience encounters every repetition and tires of it. More broadly, this pattern is particularly telling. Challengers have the strongest incentives to chase engagement, yet face the clearest diminishing returns to doing so.

Together, with the null supply relationships documented above, these results suggest that content novelty is one reason engagement incentives do not reshape issue agendas: while divisive issues are richly rewarded, the reward cannot be scaled by supplying more of it.[13]

7 Discussion

We show that congressional candidates receive more engagement when they discuss divisive national issues on Twitter. In the 2020 and 2022 election cycles, tweets referencing guns, abortion, race, Trump's 2019 impeachment, January 6th, Biden, and China received substantially larger engagement than tweets discussing the "material" issues voters care most about, such as the economy, health care, education, Covid-19, and climate. However, we find no evidence that these engagement incentives shape candidates' issue agendas. In aggregate, candidates predominantly discuss the issues voters prioritize. At the individual level, even candidates who receive especially large engagement returns from discussing divisive topics are no more likely to do so. Neither cross-candidate heterogeneity in ideology, district safety, nor incumbency status alter this conclusion.

Diminishing returns are only part of the explanation for why candidates do not chase engagement. We show that divisive topics' returns decline with frequent recent use — consistent with part of their appeal deriving from novelty and event-driven salience — but even when exhaustively used, they still generally outperform voter-prioritized issues. Other constraints must therefore be at play. One likely constraint, we propose, is representation

[13] This analysis also addresses a potential circularity concern: divisive topics may earn large boosts precisely *because* candidates rarely discuss them, so the candidates with the largest incentives to go divisive are those who use such content least. Our design varies exactly this margin — recent usage. The estimates show what happens in the instances where candidates do saturate their feeds with a high-return topic: the boost erodes. Novelty may well be part of why divisive tweets pop, but a candidate attempting to extract more engagement by reallocating their agenda toward such content would dissolve the very gains being chased, rationalizing the restraint we observe in candidates' issue agendas (see Figures 5 and 6).

and accountability: candidates tether their issue agendas to voters' priorities in order to be perceived as authentic, in step, and electorally viable. While testing this constraint directly is beyond the paper's scope, it is consistent with how some campaigns describe their own behavior (Kreiss et al. 2018).

There are, of course, limits to what we can conclude based on this analysis. Because we do not have district-level voter survey data, it is possible that voters in some districts prioritize divisive issues, but this only renders the general candidate emphasis on material topics even more striking. Researchers also need to be cautious about generalizing results from one platform to another (Larsson 2026; Stier et al. 2018), and hence it is an open question how much these patterns replicate under Elon Musk's stewardship of Twitter's successor X. Furthermore, while our topics span many of the canonical issue areas in American politics, they do not cover them all. We therefore cannot rule out that issues outside our analysis — immigration or foreign policy, for example — exhibit different patterns.

In addition, candidates may still adapt to engagement in narrower ways. Even if politicians do not reorganize the volume of tweets they send on each topic, they may respond to engagement by changing how they frame issues or which examples they foreground — a candidate who receives especially strong engagement from divisive issues may, for instance, learn to frame both divisive and material topics in more polarizing ways. Candidates may also update dynamically in ways our election-cycle-level analysis cannot detect, learning from particularly viral tweets and shifting some attention in the short run toward topics that recently performed well. Such dynamic optimization would be consistent with a few studies focusing on other national, platform, and rhetorical contexts (Frimer et al. 2023; Schöll et al. 2024; Ennser-Jedenastik et al. 2022), as well as Barbera et al. (2019), who find that members of the 113th Congress (2013-2015) adjust the issue composition of their tweets based on day-level changes in the issues their followers and the media tweet about. These narrower adaptations are certainly possible, but they are compatible with our central finding: while politicians may reframe issues or upweight topics as they become topical, their overall campaign agendas — the distribution of attention across issues — are not determined by user engagement.

More broadly, our findings suggest that concerns about social media's influence on politicians' communication strategies require nuance. The common refrain that engagement incentives encourage politicians to adopt more polarizing rhetoric (e.g. Ballard et al. 2022 or Heseltine and Dorsey 2022) does not extend to issue agendas: while candidates may occasionally benefit from invoking divisive topics, their overall allocation of attention appears largely independent of the returns to engagement. Polarizing topics do receive greater engagement on Twitter, likely shaping the issues users see via algorithmic amplification — but

this amplification is driven by user demand, not candidate supply.

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9 Disclosure Statement

The authors report there are no competing interests to declare.

10 Biographical Notes

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11 Data Availability Statement

All data required to replicate our results are available upon request.

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12 Appendix

A Topic Dictionaries

We use the R package `quanteda` to label tweets based on the following dictionaries. If a tweet contains any keyword in the dictionary, it is labeled as that topic. The * symbols indicate a catch-all for any characters. For example, our keyword 'covid*' would catch the term 'covid-19' in a tweet. This functionality is particularly important for catching diverse hashtags that contain key terms.

Table A.1: Topic Dictionaries

Topic	**Dictionary**
Trumpism	make america great again, keep america great, kag, maga, antifa, china virus, *wuhanflu*, wuhan virus, election integrity, stop the steal,*hydroxycholoroquine*, fraudulent vote*, election fraud, voter fraud, fraudelent voting, fraudulent election, radical left*, radical democrats, patriot, rino, fake news, americafirst, america first, save america, stolen election, rigged election, build the wall, drain the swamp, back the blue, blue lives matter
Leftism	defund the police, systemic racism, green new deal, black lives matter, blacktranslivesmatter, intersectional, jim crow, white supremacy, white nationalism, racial justice, people of color, women of color, wealth tax, indigenous, black and brown, *incarcerat*, *carceral*, transphobia, transphobic, breathe act, medicare for all, latinx, womxn, *lgbtq*, stop asian hate, stop aapi hate, *patriarchy*, tax the rich, *for profit prison*, *private prison*, prison abolition, *oppress*, top one percent*, voter suppression, marginalized group*, marginalized community
Incivility	*fuck*,*shit*, *bitch*, damn, damnit, goddamnit, *idiot*, *asshole*, *stupid*, dumb, *dumbass*, *moron*, *retard*, fool, ass, *bastard*, wtf, impeachthemf, mf, *libtard*, crap
Trump	*trump*
Biden	*biden*

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Table A.1 (continued)

Topic	Dictionary
Guns	gun, *gunviolence*, pistol, *shooting*, background check*, *background-check*, hr8, *uvalde*, moms demand, march for our lives, parkland, second amendment, bump stock*, ak47, assualt rifle*, columbine, *banassaultrifle*
Abortion	*abortion*, *pro life*, *pro choice*, *fetus*, *born alive*, *infanticide*, *planned parenthood*, *reproductive right*, *unborn*
Impeachment	*impeach*, *inquiry*, *whistleblower*, vindman, obstruction of congress
January 6th	*impeach*, *jan 6* , *january 6*, *insurrection*, capitol police, coup, overturn the election, *elector, *electors, *capitol riot*, storm the capitol, capitol assault, capitol attack
China	*wuhan*, china, ccp, hong kong, taiwan, hongkong, president xi, jin ping, beijing, chinese, tibet, axinjiang, uyghur
Race	black lives matter, blacktranslivesmatters, *police*, all lives matter, george floyd, *policing*, back the blue, blue lives matter, *racism*, *racist*, aapi, asian american*,whites, white people, white america*,*minorities*, african american*, breonna taylor, eric garner, philando castille, freddie gray, tamir rice, michael brown, *cops*, *saytheirnames*, derek chauvin, people of color, poc,jim crow, white supremacy, white nationalism, racial, woc, women of color, hispanic*, latino*, latina*, latinx, indian american*, stop asian hate, stop aapi hate, law enforcement, indigenous, native american*, black, *blackwomen*, *blackcaucus*, *blackgirl*, *blackboy*, communities of color
COVID-19	*covid*, *coronavirus*, *pandemic*, *virus*,*vaccine*, *wuhan*, *pfizer*, *moderna*, *astrazeneca*, *fauci*, mask, masks, pcr, rapid test, rapid testing
Climate	*climate*, *global warming*, *emissions*, clean energy, *environment*, fossil fuel*, *carbon*, renewable energy, green new deal, green bad deal, green economy, *greenhouse gas*, green energy, green infrastructure, green jobs, green coalition, green ferries, *planet*, *ozone*
Education	schools, *education*, critical race theory, colleges, universities, prek, student*, kindergarten, teacher*, classroom, tuition

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Table A.1 (continued)

Topic	**Dictionary**
Economy	*economy*, *economic*, tax*, *inflation*, business*, *worker*, stock market, free market*, middle class, jobs, job creation, wage, *minimumwage*, *livablewage*, financial, finance, welfare*, social security, paid leave, maternity leave, paternity leave, ubi, ppp, paycheck protection program, salary, income*, corporate, corporation*, pension*, national debt, budget, deficit, employment, employees, employer*, labor, unions, union member*, union fight*, union bargain*, union money, union dues, union job*, union leader*, unionize, union president, trade policy, trade dispute*, trade agreement*
Health	*health*, *health care*, health service*, health program*, mental health*, affordable care act, *medicare*, *medicaid*, obama care, *trumpcare*, *opioid*, *pharma*, insulin, medicine, medical, medication, *prescription*, preexisting condition*

B Distribution of Topics

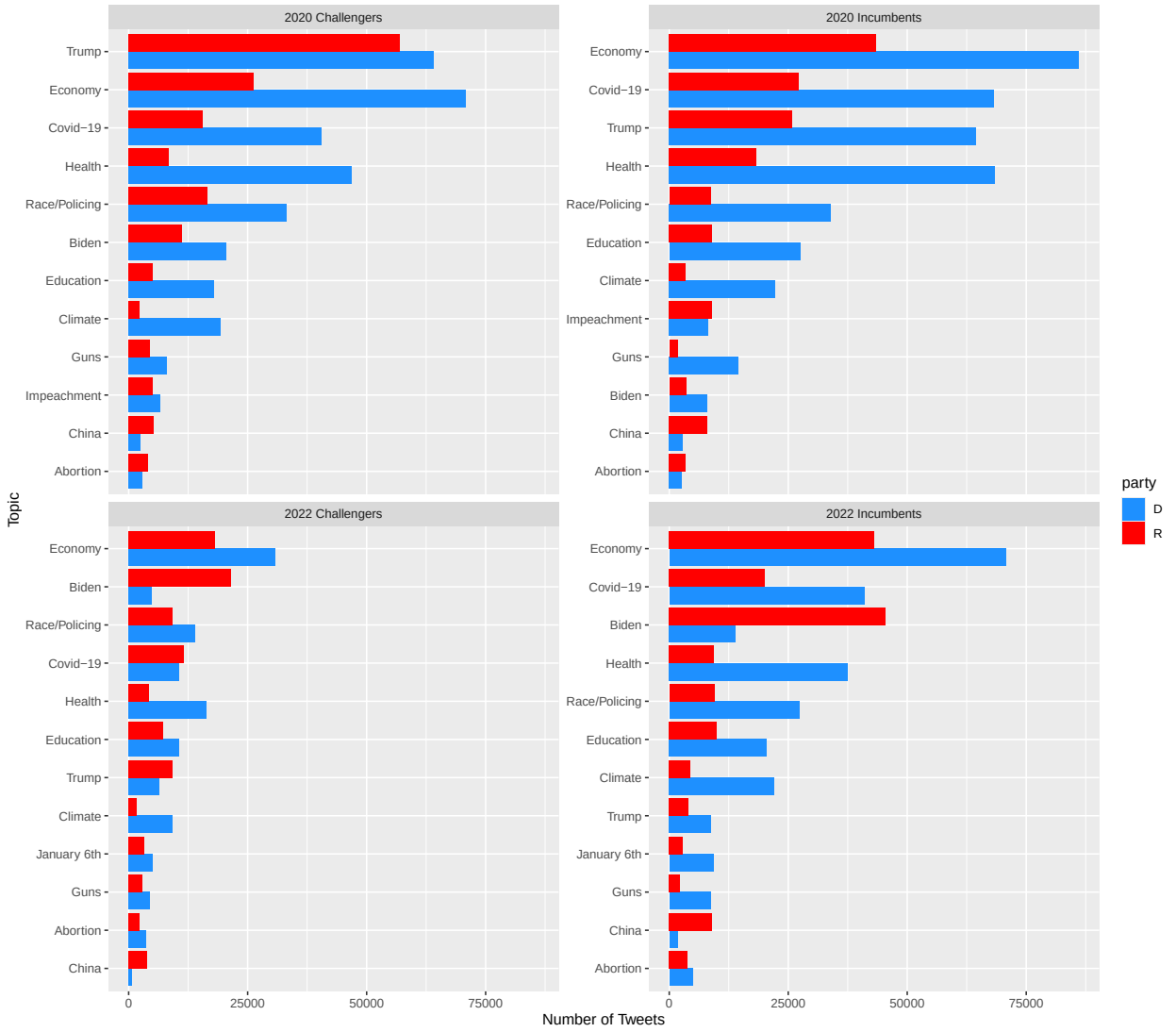


Figure A.1: Distribution of Topics by Party, Election Cycle, and Challenger vs. Incumbent status. The bars show the number of tweets sent mentioning each topic in the 2020 and 2022 election cycles. Within each pane, topics are ordered top-down by the average of the two parties' bars, mirroring Figure 5 in the main text; the tone categories (Trumpism, Leftism, and incivility) are omitted. In our data, the 2020 election cycle spans from January 2019-December 2020, and the 2022 election cycle spans from January 2021-May 2022.

C Candidate-Level Heterogeneity in Engagement Incentives

To test whether engagement returns vary with candidate ideology or district safety, we re-estimate our main specification (Equation 2) within each party × candidate-type × cycle subsample, interacting each topic indicator with a candidate-level moderator M_c. For tweet w by candidate c,

$$Y_w = \beta_0 + \sum_{j=1}^{12} \beta_j T_w^{(j)} + \sum_{j=1}^{12} \gamma_j \left(T_w^{(j)} \times M_c \right) + \boldsymbol{\eta}\mathbf{X} + \tilde{\boldsymbol{\eta}} \left(\mathbf{X} \times M_c \right) + \alpha_c + \delta_t + \varepsilon_w \quad (5)$$

where M_c is candidate ideology — first-dimension DW-Nominate for incumbents and DW-DIME (Bonica 2018) for challengers, demeaned and standardized among the candidates in each subsample — or, for the district-safety models, incumbents' general-election vote share demeaned with respect to 50 percent. The moderator's main effect is absorbed by the candidate fixed effects α_c, so the objects of interest are the interactions γ_j, which we plot below. For the ideology models, each γ_j is the change in topic j's (log-scale) engagement effect associated with a one standard deviation increase in ideology, i.e., in the conservative direction. A positive interaction for Democrats therefore means a topic's engagement effect is larger for moderates; for Republicans, a *negative* interaction means larger effects for moderates. For the district-safety models, each γ_j is the change in topic j's effect per additional percentage point of vote share.

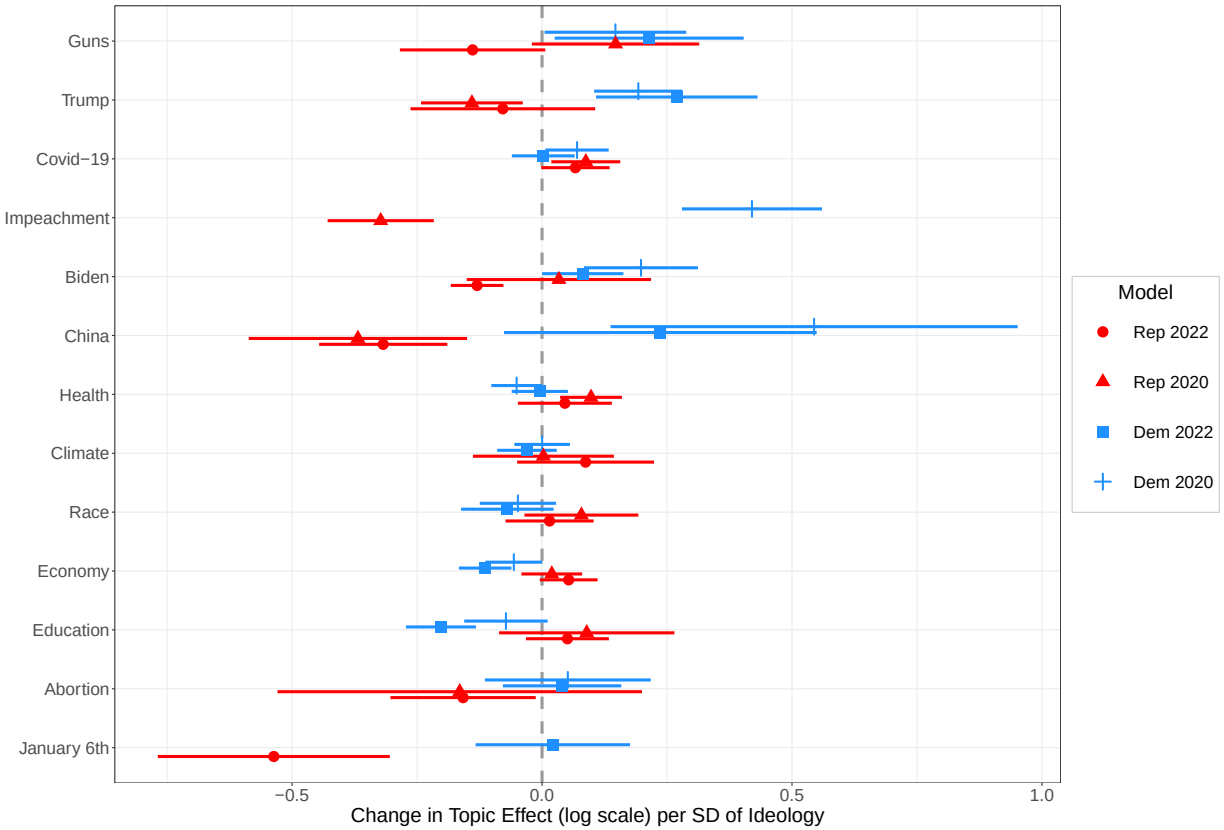


Figure A.2: Heterogeneity in Incumbents' Engagement Returns by Ideology. Points are topic × ideology interaction coefficients from four separate models split by party and election cycle, where ideology is standardized first-dimension DW-Nominate. Positive Democratic and negative Republican interactions both indicate larger topic effects for ideological moderates. Whiskers are 95% confidence intervals from candidate-clustered standard errors.

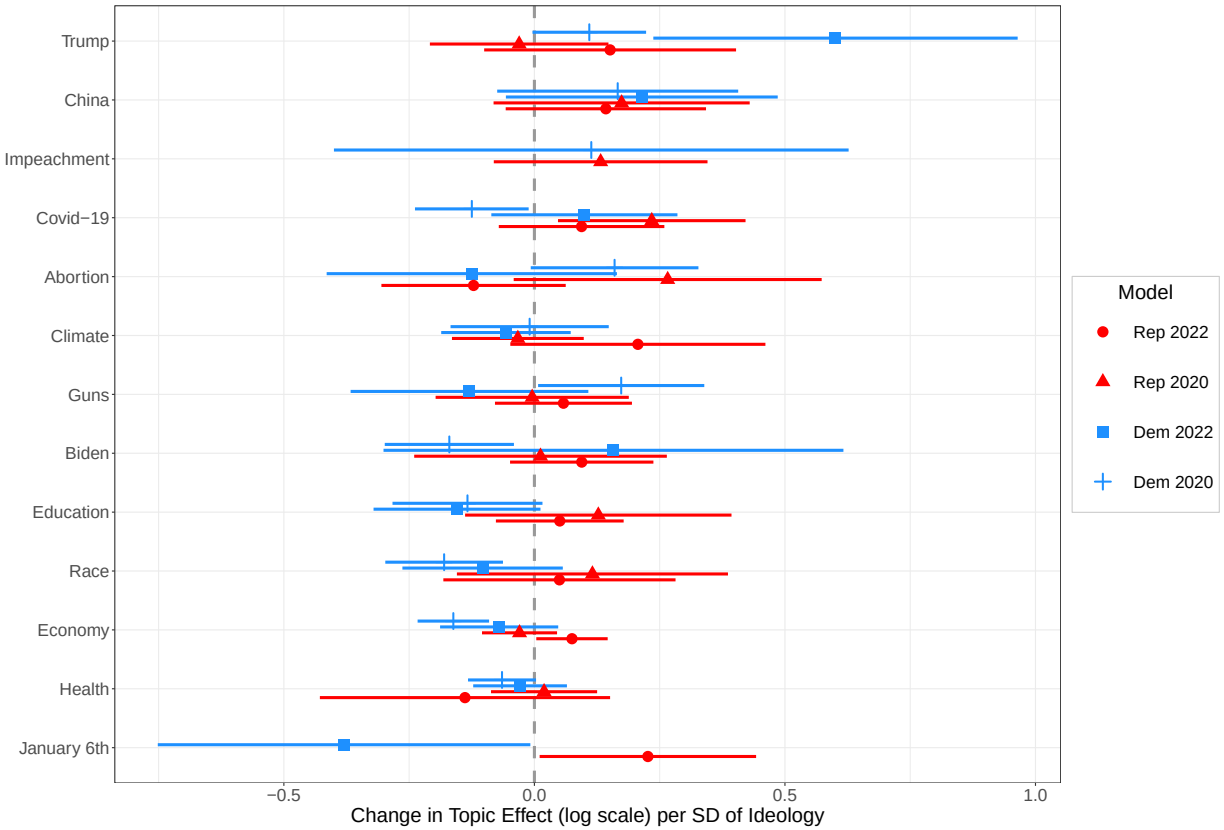


Figure A.3: Heterogeneity in Challengers' Engagement Returns by Ideology. Points are topic × ideology interaction coefficients from four separate models split by party and election cycle, where ideology is standardized DW-DIME (Bonica 2018). Positive Democratic and negative Republican interactions both indicate larger topic effects for ideological moderates. Whiskers are 95% confidence intervals from candidate-clustered standard errors.

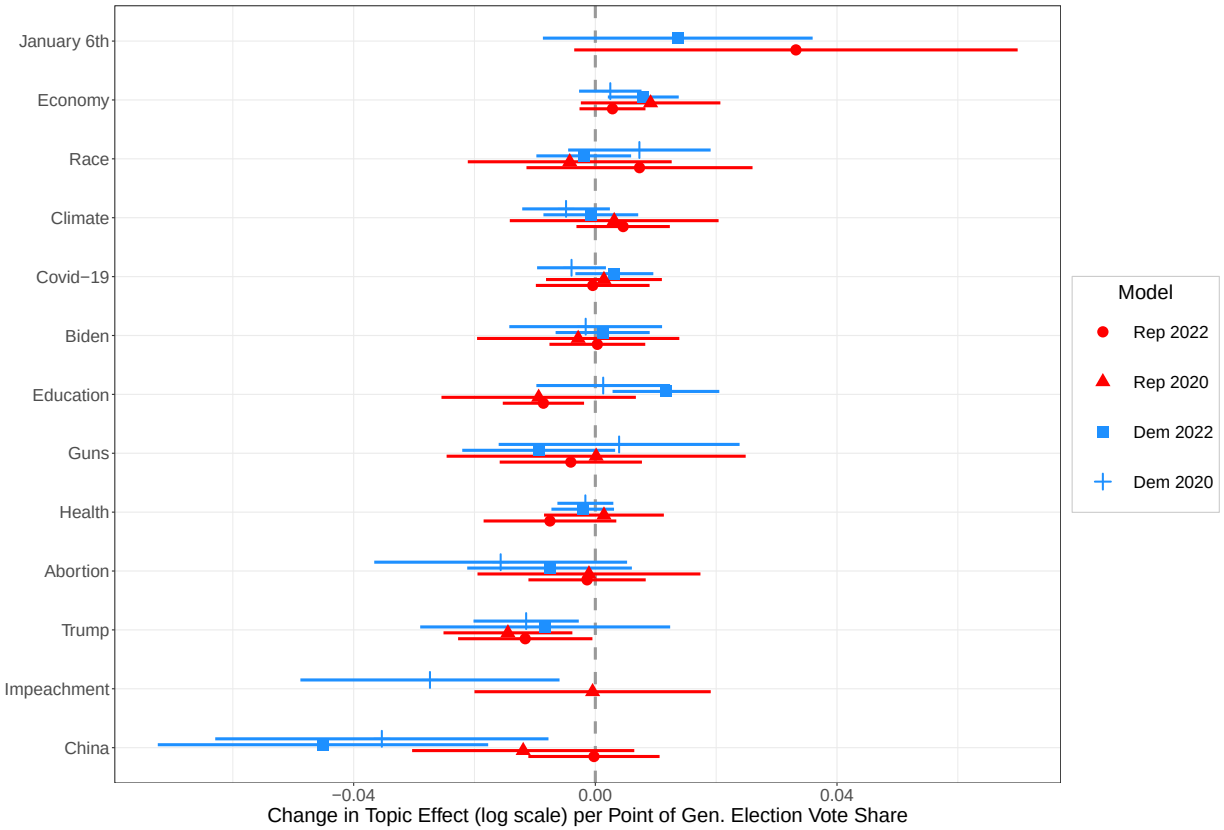


Figure A.4: Heterogeneity in Incumbents' Engagement Returns by District Safety. Points are topic × vote-share interaction coefficients from four separate models split by party and election cycle, where general-election vote share is centered at 50 percent. Each coefficient is the change in a topic's (log-scale) engagement effect per percentage point of vote share. Whiskers are 95% confidence intervals from candidate-clustered standard errors.

Three patterns emerge. First, for district safety (Figure A.4), nearly all interactions are null and small in magnitude: engagement returns are essentially constant across safe and competitive districts. Second, for challengers (Figure A.3), the ideology interactions are mostly null, with no pattern consistent across parties or cycles. Third, among incumbents (Figure A.2), a handful of divisive topics — China, impeachment, guns (Democrats only), and January 6th (Republicans only) — show significantly larger effects for moderates of both parties (positive interactions for Democrats, negative for Republicans). In addition, among Democratic incumbents, the economy and education interactions are slightly negative, indicating that moderates fare *worse* on these material issues. But in most models, particularly in those testing issues voters prioritize most, engagement returns are comparable across the ideological spectrum.

D Candidate-Level Heterogeneity in Issue Agendas

D.1 By Ideology

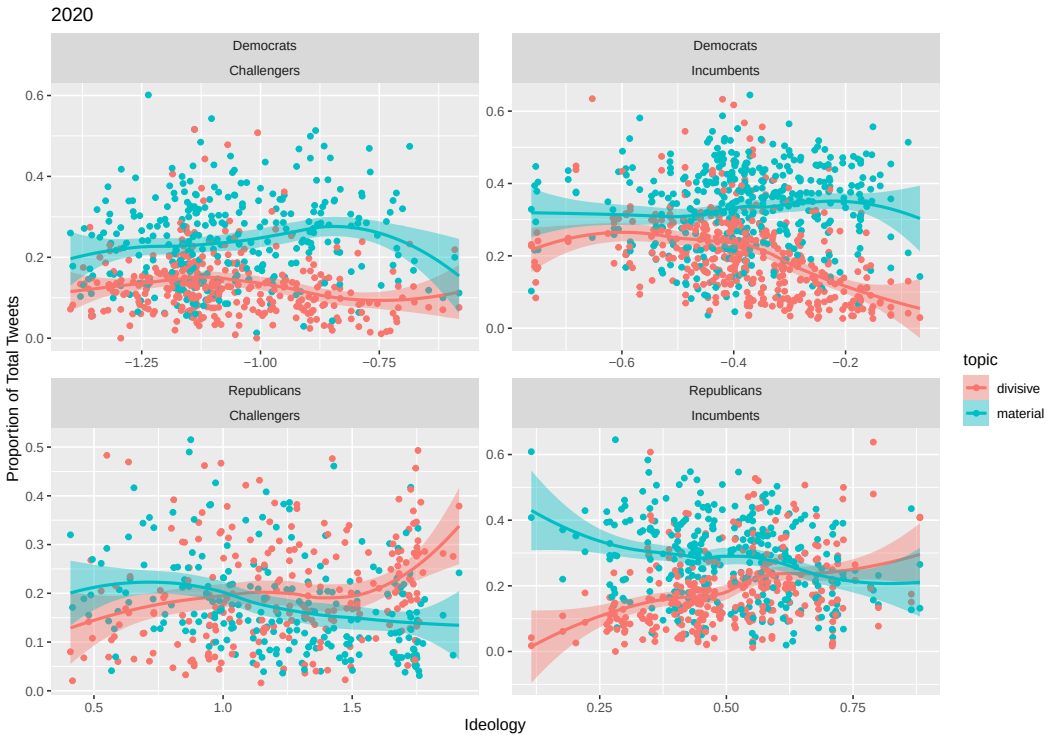


Figure A.5: 2020 Election Cycle: Ideological Extremists Among Incumbents Only are More Likely to Discuss Divisive Issues. Each dot is the share of a candidate's total tweets sent on that issue-type. First-dimension ideology is measured with DW-Nominate for incumbents and DW-DIME (Bonica 2018) for challengers. Curves are LOESS smoothers.



Figure A.6: 2022 Election Cycle: Ideological Extremists Among Incumbents Only are More Likely to Discuss Divisive Issues. Each dot is the share of a candidate's total tweets sent on that issue-type. First-dimension ideology is measured with DW-Nominate for incumbents and DW-DIME (Bonica 2018) for challengers. Curves are LOESS smoothers.

D.2 By District Safety



Figure A.7: Incumbents in Safe Districts are More Likely to Discuss Divisive Issues. Each dot is the share of a candidate's total tweets sent on that issue-type. FIgure titles denote the election cycle. Curves are LOESS smoothers.

E Voter Priorities and Engagement Incentives

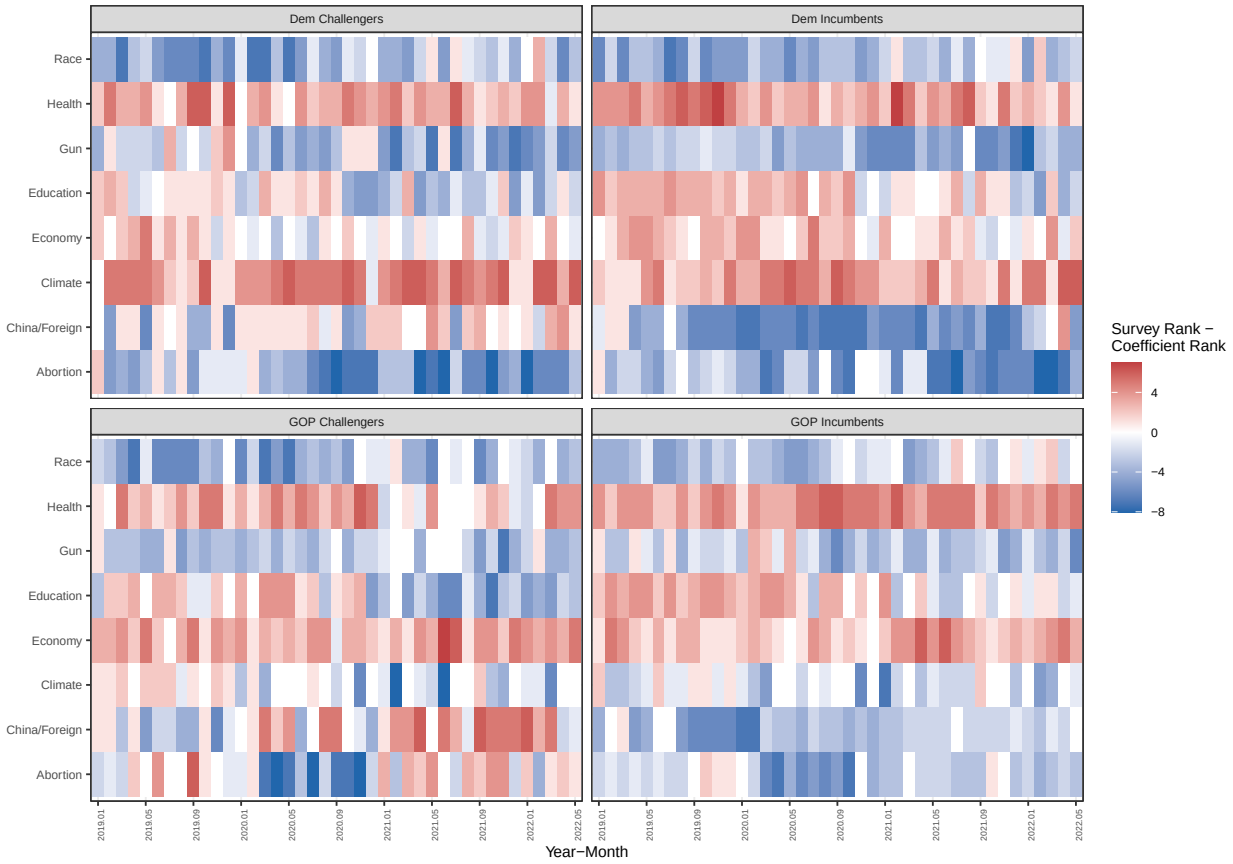


Figure A.8: Voters' Issue Priorities and Users' Engagement Priorities Diverge Within Each Month. Each cell is an issue's YouGov importance rank among the party's voters minus the rank of the issue's engagement coefficient in that month, where both rankings run from 1 (least important or engaged-with) to 8 (most). Coefficients obtained by estimating specification 2 at the month-level (sans δ_t). Red cells indicate voters prioritize the issue more than users engage with it; blue cells indicate the reverse; white cells indicate the two audiences rank the issue similarly.

Here we compare voters' issue priorities to Twitter users' engagement priorities at the month level, allowing both to shift over the election cycle.

We first estimate 2 (sans the month fixed effects δ_t) within each month to obtain month-specific topic-engagement associations. We then identify the set of issues included in both our models and YouGov's monthly surveys, which ask respondents which issue is most important to them; these are: abortion, race, guns, china/foreign policy, climate, education, health, and the economy. Within each month and party, we rank these eight issues twice: once by the share of the party's voters naming the issue most important (from YouGov), and once

by the issue's engagement coefficient (from our models), where 1 is the least and 8 the most important or engaged-with issue. Each cell of Figure A.8 reports the issue's survey rank minus its coefficient rank in that month. If engagement incentives tracked voters' priorities, the heatmap would be white throughout, with individual cells deviating only transiently.

Instead, even when we allow voters' priorities to shift over time and compare them to engagement incentives within each month, the two consistently diverge. Consistent with the discussion in Section 6.1, voters of both parties durably prioritize the economy, health care, and education — rows that stay red across nearly the entire study period — while users spurn these issues in favor of issues less valued by voters at the time, such as race, guns, and abortion, whose rows generally stay blue. In addition, while Democratic voters deeply prioritize climate, typically sitting at the second or first most important issue to these voters throughout the period of study (see Figure 1), Democratic candidates' users engage considerably less with climate relative to other issues.

F Diminishing Returns to Repetition: 2022 Election Cycle

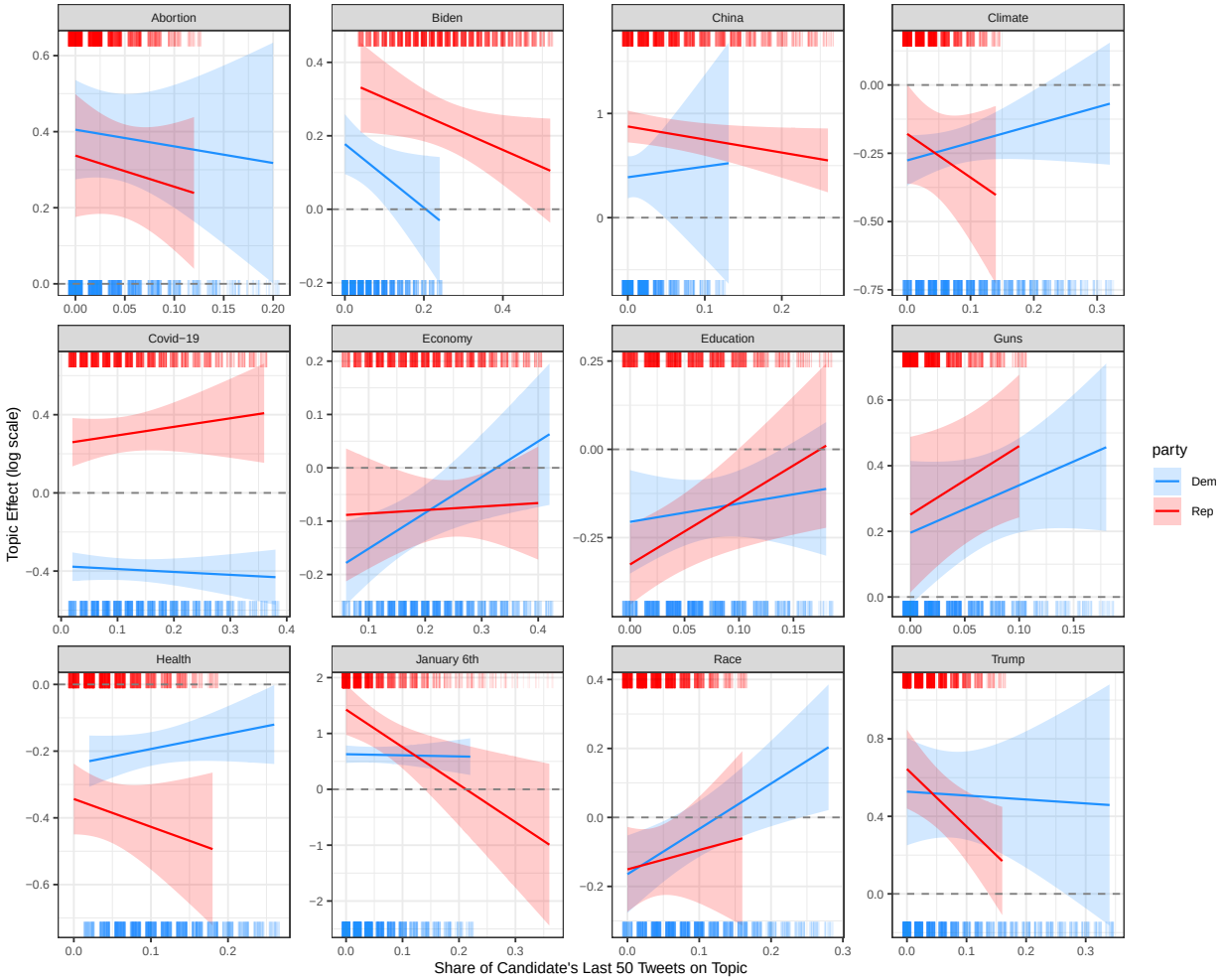


Figure A.9: Diminishing Returns to Repetition Among 2022 Incumbents. Lines plot the implied effect of each topic ($\hat{\beta}_j + \hat{\gamma}_j S$, Equation 4) as a function of the share of the candidate's 50 preceding tweets mentioning that topic, over the central 90% of the observed intensity distribution among the topic's tweets. Bands are 95% confidence intervals from candidate-clustered standard errors. Rug marks show the observed intensity distributions (Democrats on the bottom edge, Republicans on the top edge).

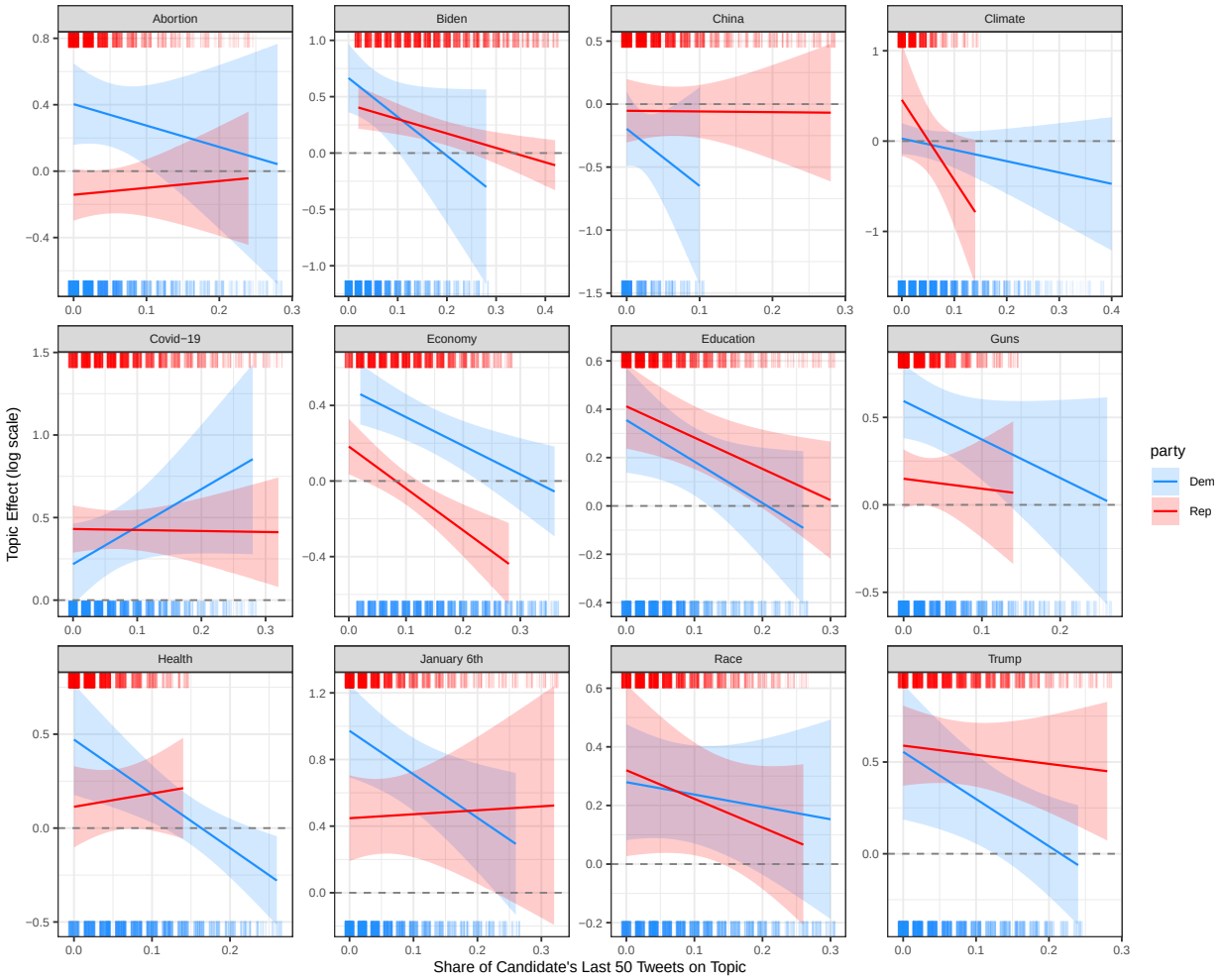


Figure A.10: Diminishing Returns to Repetition Among 2022 Challengers. Lines plot the implied effect of each topic ($\hat{\beta}_j + \hat{\gamma}_j S$, Equation 4) as a function of the share of the candidate's 50 preceding tweets mentioning that topic, over the central 90% of the observed intensity distribution among the topic's tweets. Bands are 95% confidence intervals from candidate-clustered standard errors. Rug marks show the observed intensity distributions (Democrats on the bottom edge, Republicans on the top edge).

G Additional Information for Main Topic-Engagement Models

G.1 Plotting Coefficients on Percent-Change Scale

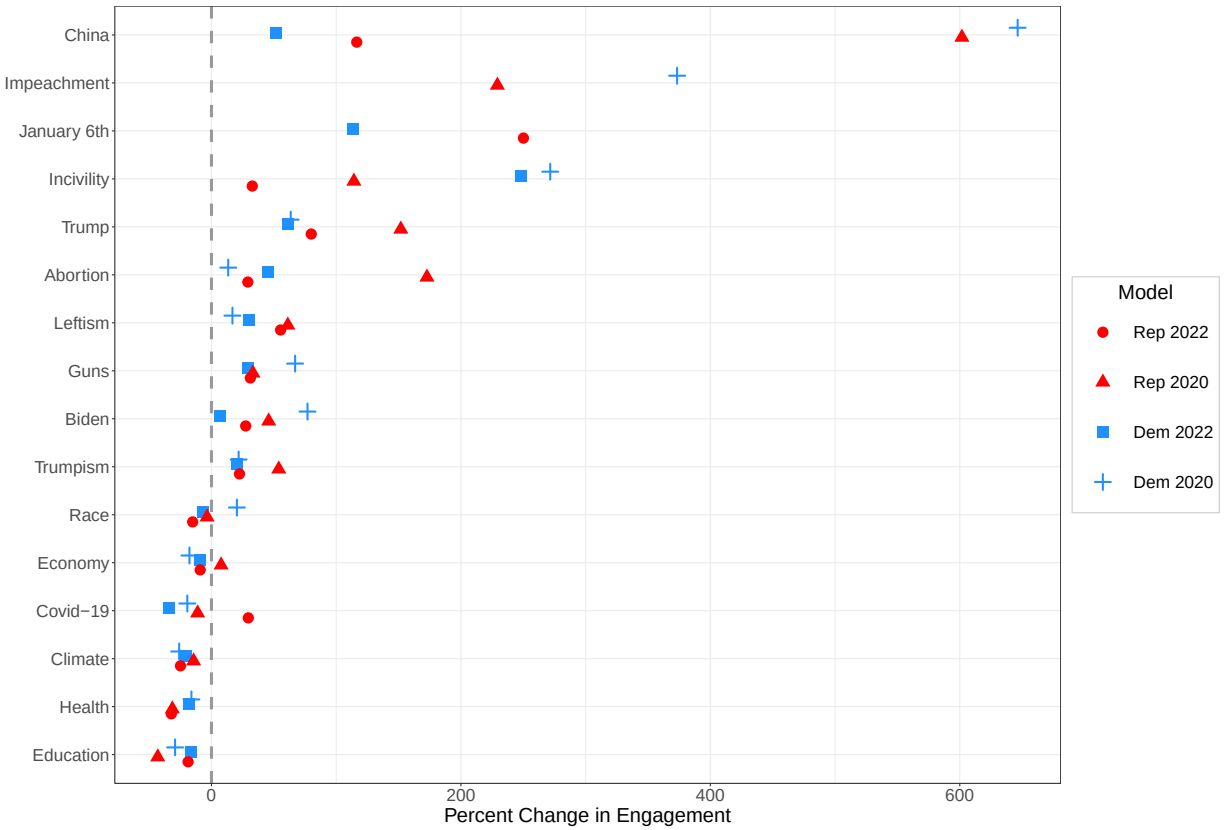


Figure A.11: Main-Text Incumbent Estimates on the Percent-Change Scale. Point estimates from the main-text negative-binomial models for incumbents (Figure 3), transformed to $(e^{\beta_j} - 1) \times 100$: the within-candidate, within-month percent change in engagement associated with discussing topic j. Confidence intervals are reported on the coefficient scale in Figure 3 and Table A.2.

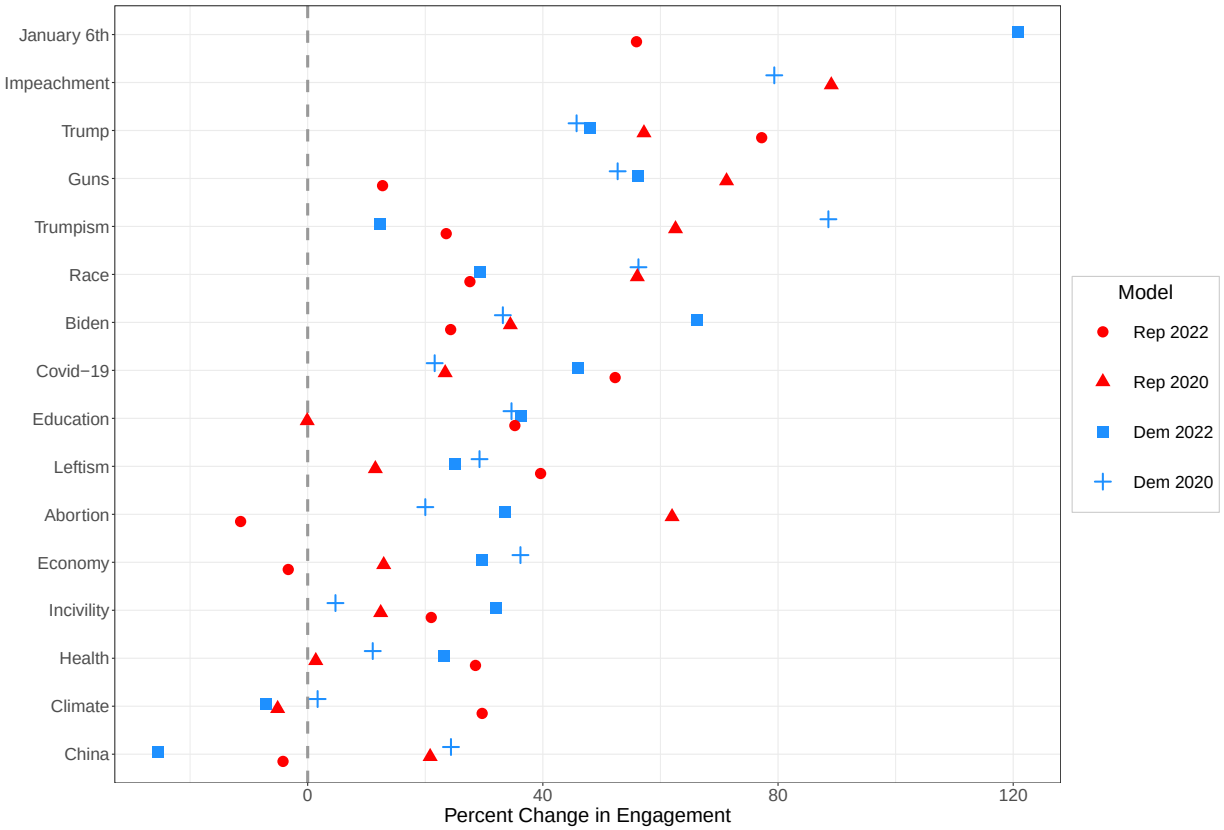


Figure A.12: Main-Text Challenger Estimates on the Percent-Change Scale. Point estimates from the main-text negative-binomial models for challengers (Figure 4), transformed to $(e^{\beta_j} - 1) \times 100$: the within-candidate, within-month percent change in engagement associated with discussing topic j. Confidence intervals are reported on the coefficient scale in Figure 4 and Table A.3.

G.2 Main Text Regression Results in Table Format

Table A.2: Association between Topics and Engagement for Incumbents

	Democrats		Republicans	
	2020	2022	2020	2022
Trump	0.49 (0.04)	0.48 (0.1)	0.92 (0.07)	0.59 (0.09)
Guns	0.51 (0.12)	0.26 (0.08)	0.29 (0.1)	0.27 (0.1)
Abortion	0.13 (0.14)	0.37 (0.06)	1 (0.14)	0.26 (0.06)
Impeachment	1.55 (0.11)		1.19 (0.12)	
January 6th		0.76 (0.08)		1.25 (0.16)
Race	0.19 (0.07)	-0.07 (0.04)	-0.04 (0.07)	-0.16 (0.06)
Biden	0.57 (0.07)	0.07 (0.04)	0.38 (0.13)	0.24 (0.04)
China	2.01 (0.25)	0.42 (0.16)	1.95 (0.12)	0.77 (0.08)
Covid-19	-0.22 (0.04)	-0.42 (0.03)	-0.12 (0.05)	0.26 (0.05)
Climate	-0.3 (0.05)	-0.23 (0.04)	-0.15 (0.08)	-0.29 (0.07)
Education	-0.35 (0.06)	-0.18 (0.04)	-0.56 (0.07)	-0.21 (0.06)
Economy	-0.19 (0.03)	-0.1 (0.03)	0.07 (0.04)	-0.09 (0.03)
Health	-0.18 (0.03)	-0.2 (0.03)	-0.38 (0.04)	-0.39 (0.05)
Trumpism	0.2 (0.09)	0.18 (0.14)	0.43 (0.07)	0.2 (0.07)
Leftism	0.16 (0.06)	0.27 (0.05)	0.48 (0.1)	0.44 (0.08)
Incivility	1.31 (0.53)	1.25 (0.49)	0.76 (0.23)	0.28 (0.1)
Obs	588342	400384	286081	268229
Pseudo R2	0.11	0.1	0.1	0.09

[1] Clustered standard errors at user-level shown in parentheses.

[2] $(e^{\beta} - 1) * 100$ gives percent change in engagement from mentioning a topic

Table A.3: Association between Topics and Engagement for Challengers

	Democrats		Republicans	
	2020	2022	2020	2022
Trump	0.38 (0.07)	0.39 (0.13)	0.45 (0.06)	0.57 (0.09)
Guns	0.42 (0.09)	0.45 (0.1)	0.54 (0.16)	0.12 (0.06)
Abortion	0.18 (0.07)	0.29 (0.11)	0.48 (0.12)	-0.12 (0.06)
Impeachment	0.58 (0.18)		0.64 (0.09)	
January 6th		0.79 (0.12)		0.44 (0.12)
Race	0.45 (0.07)	0.26 (0.09)	0.45 (0.09)	0.24 (0.11)
Biden	0.29 (0.11)	0.51 (0.15)	0.3 (0.08)	0.22 (0.06)
China	0.22 (0.24)	-0.29 (0.11)	0.19 (0.12)	-0.04 (0.1)
Covid-19	0.2 (0.09)	0.38 (0.1)	0.21 (0.09)	0.42 (0.07)
Climate	0.02 (0.08)	-0.07 (0.09)	-0.05 (0.08)	0.26 (0.22)
Education	0.3 (0.13)	0.31 (0.09)	0 (0.11)	0.3 (0.07)
Economy	0.31 (0.05)	0.26 (0.06)	0.12 (0.04)	-0.03 (0.04)
Health	0.1 (0.04)	0.21 (0.09)	0.01 (0.06)	0.25 (0.12)
Trumpism	0.63 (0.22)	0.12 (0.1)	0.49 (0.07)	0.21 (0.07)
Leftism	0.26 (0.08)	0.22 (0.09)	0.11 (0.1)	0.33 (0.21)
Incivility	0.05 (0.1)	0.28 (0.17)	0.12 (0.1)	0.19 (0.13)
Obs	816283	314759	409074	256060
Pseudo R2	0.09	0.09	0.09	0.11

[1] Clustered standard errors at user-level shown in parentheses.

[2] $(e^{\beta} - 1) * 100$ gives percent change in engagement from mentioning a topic

H Robustness Checks for Main Topic-Engagement Models

H.1 Using OLS Models

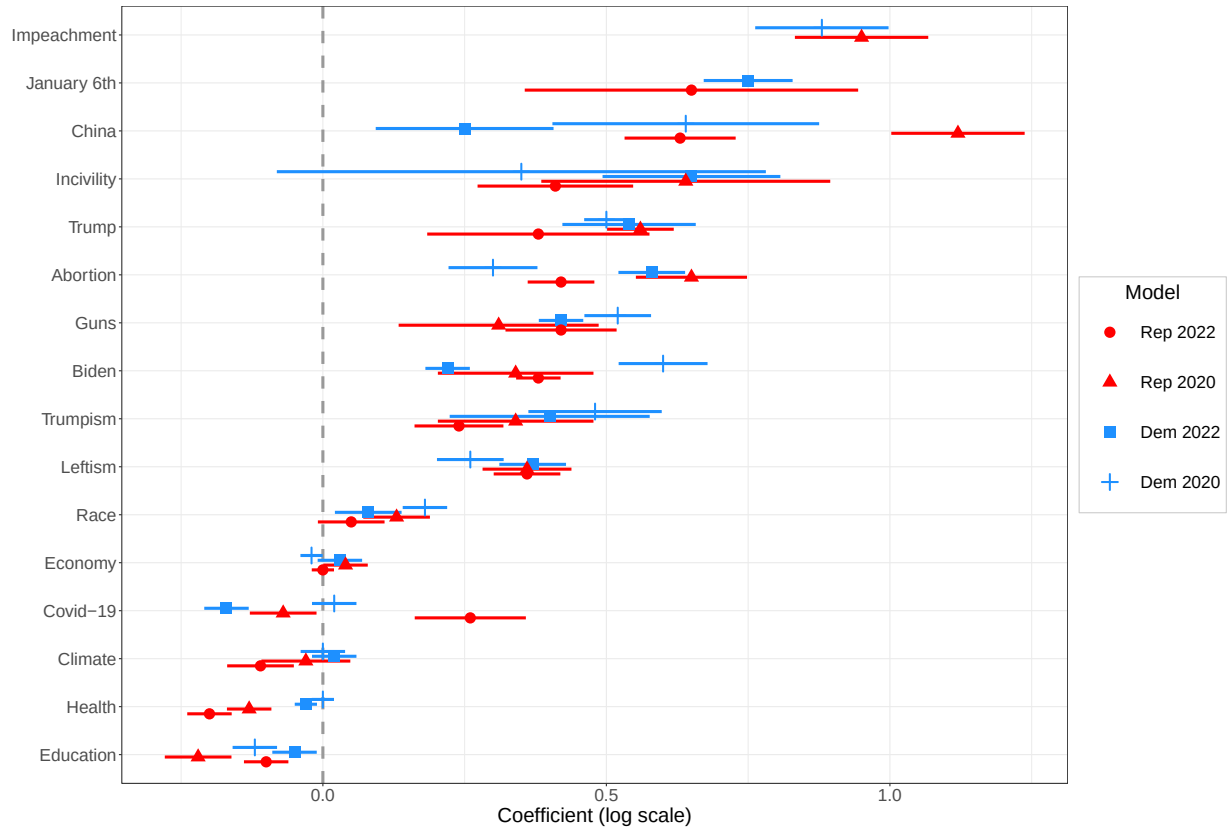


Figure A.13: Association between Topics and Engagement for Incumbents, Estimated with OLS. Coefficients from the specification in Equation 2 estimated by OLS with $\log(1 + \text{engagement})$ as the outcome. Whiskers are 95% confidence intervals from candidate-clustered standard errors. The topic ordering closely tracks the negative-binomial estimates reported in Figure 3.

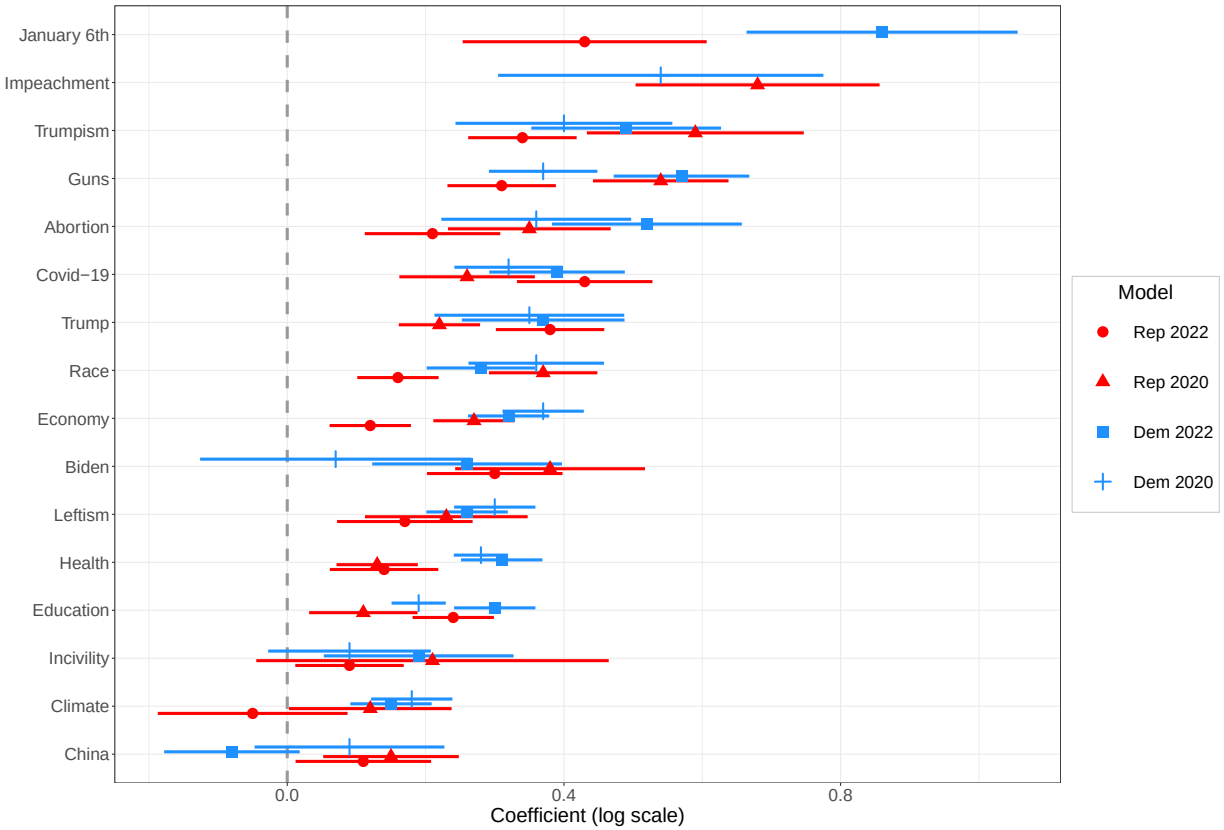


Figure A.14: Association between Topics and Engagement for Challengers, Estimated with OLS. Coefficients from the specification in Equation 2 estimated by OLS with $\log(1 + \text{engagement})$ as the outcome. Whiskers are 95% confidence intervals from candidate-clustered standard errors. The topic ordering closely tracks the negative-binomial estimates reported in Figure 4.

H.2 Modeling Candidates with Random Effects

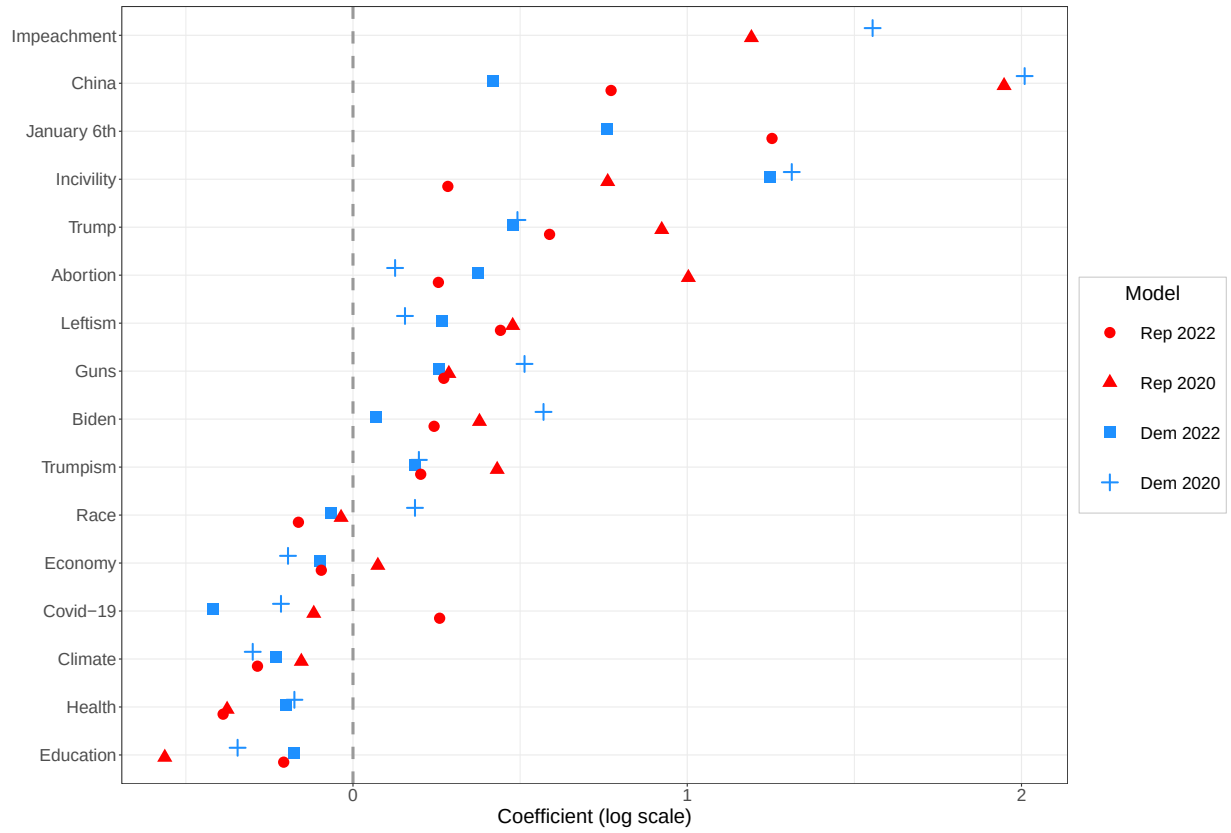


Figure A.15: Association between Topics and Engagement for Incumbents, with Candidate Random Effects. Point estimates from re-estimating the main negative-binomial models with candidate random intercepts in place of candidate fixed effects (month fixed effects retained). Across all topic coefficients (incumbents and challengers pooled), the maximum absolute deviation from the fixed-effects estimates in the main text is 0.001 (correlation = 1.00). Standard errors are not displayed: cluster-robust standard errors are not currently supported for mixed-effects negative-binomial models in R, and the model-based standard errors are severely downward biased in the presence of within-candidate dependence. The purpose of this figure is to show that the choice between fixed and random candidate effects has no effect on the point estimates.

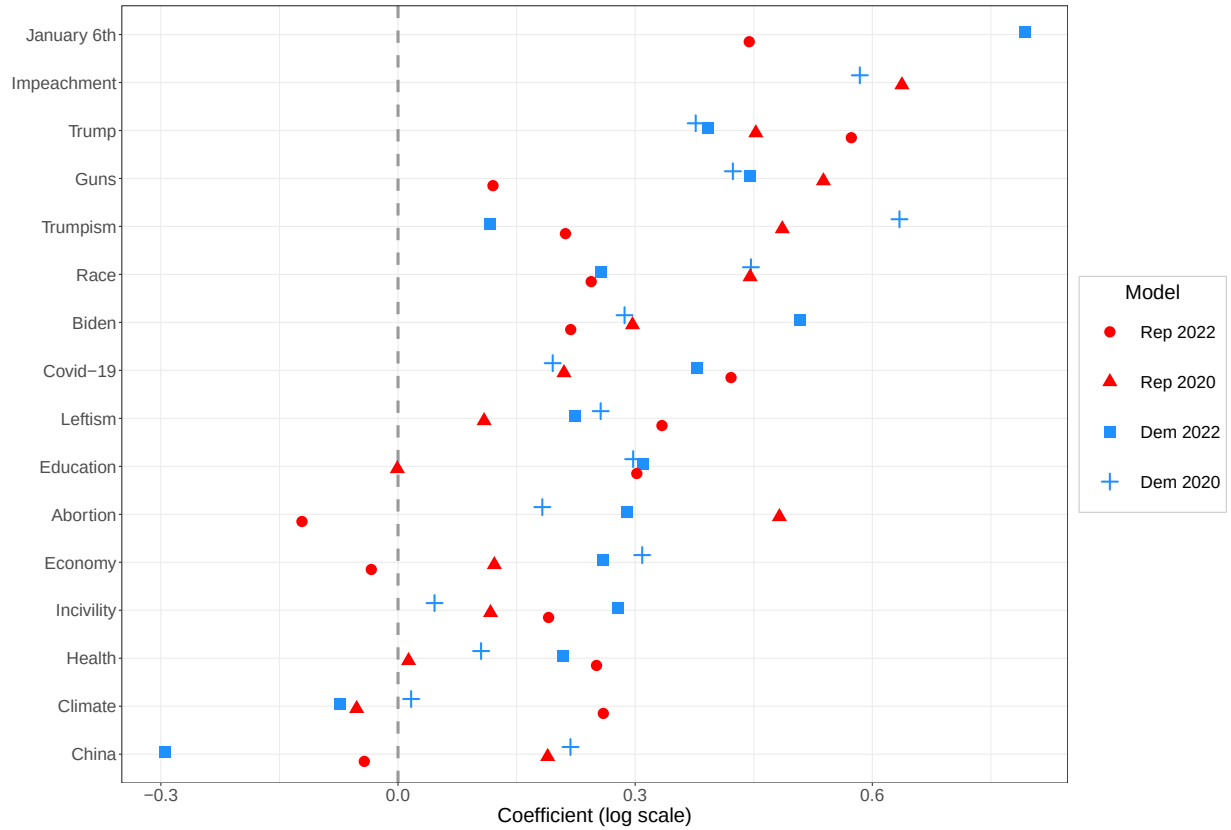


Figure A.16: Association between Topics and Engagement for Challengers, with Candidate Random Effects. Point estimates from re-estimating the main negative-binomial models with candidate random intercepts in place of candidate fixed effects (month fixed effects retained). Across all topic coefficients (incumbents and challengers pooled), the maximum absolute deviation from the fixed-effects estimates in the main text is 0.001 (correlation = 1.00). Standard errors are not displayed: cluster-robust standard errors are not currently supported for mixed-effects negative-binomial models in R, and the model-based standard errors are severely downward biased in the presence of within-candidate dependence. The purpose of this figure is to show that the choice between fixed and random candidate effects has no effect on the point estimates.

H.3 Modeling Likes and Retweets Separately

Figures A.17–A.20 re-estimate the main models with likes and retweets as separate outcomes. The three sets of estimates are highly correlated: across all 120 topic coefficients (on the log scale), the main-text estimates correlate at 0.99 with the likes-only estimates and 0.98 with the retweets-only estimates, and the likes- and retweets-only estimates correlate at 0.95 with each other. The only coefficient with a meaningful difference between the likes- and retweets-only models is China, whose retweet coefficients consistently exceed its like coefficients. This indicates that the massive engagement gains associated with discussing

China for incumbents in 2020 were driven disproportionately by resharing.

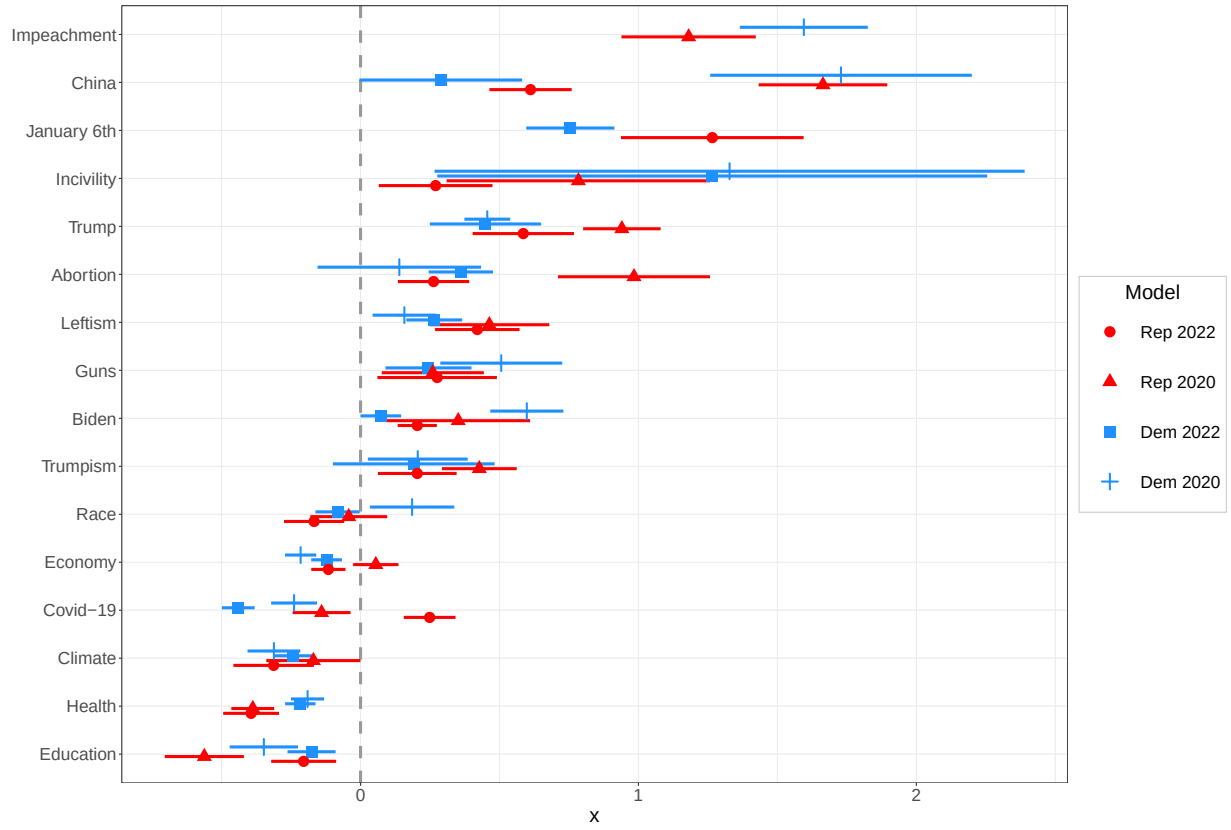


Figure A.17: Association between Topics and Likes for Incumbents. Coefficients from re-estimating the main negative-binomial models with likes (rather than total engagement) as the outcome. Whiskers are 95% confidence intervals from candidate-clustered standard errors.

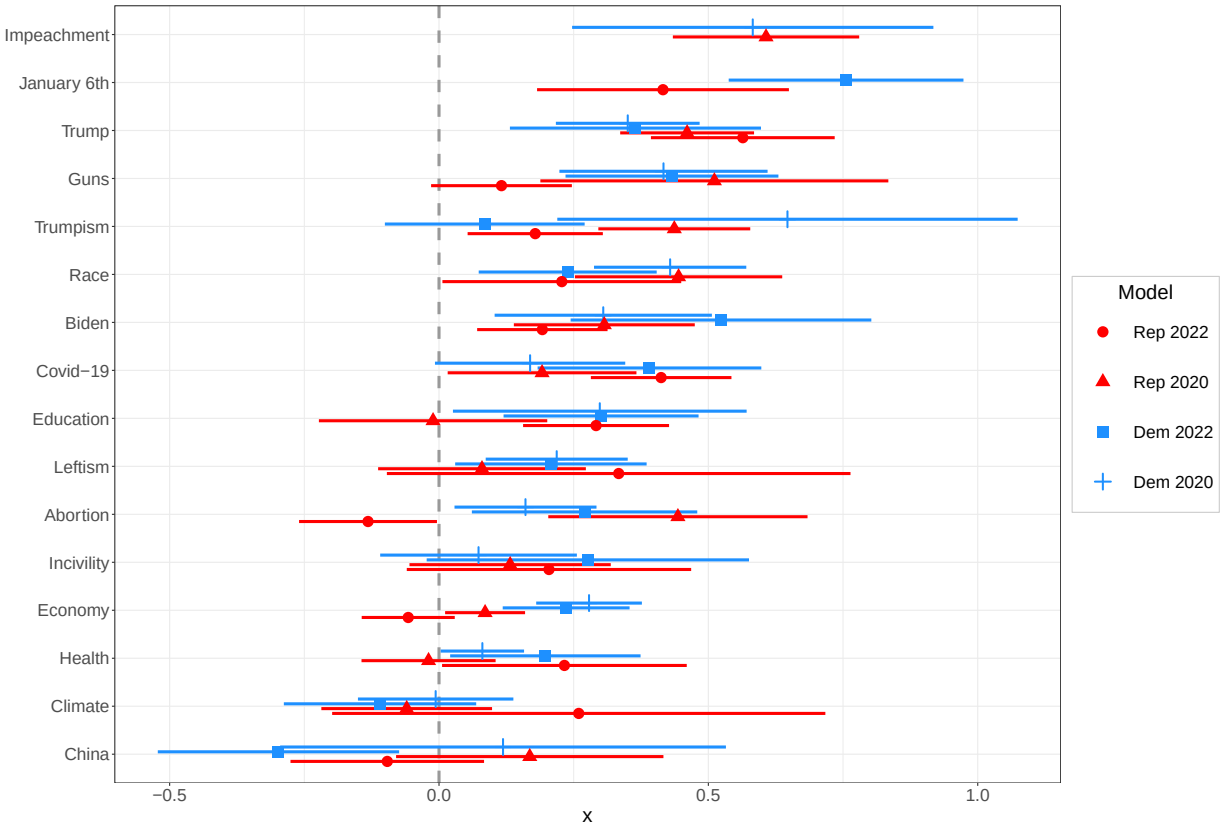


Figure A.18: Association between Topics and Likes for Challengers. Coefficients from re-estimating the main negative-binomial models with likes (rather than total engagement) as the outcome. Whiskers are 95% confidence intervals from candidate-clustered standard errors.

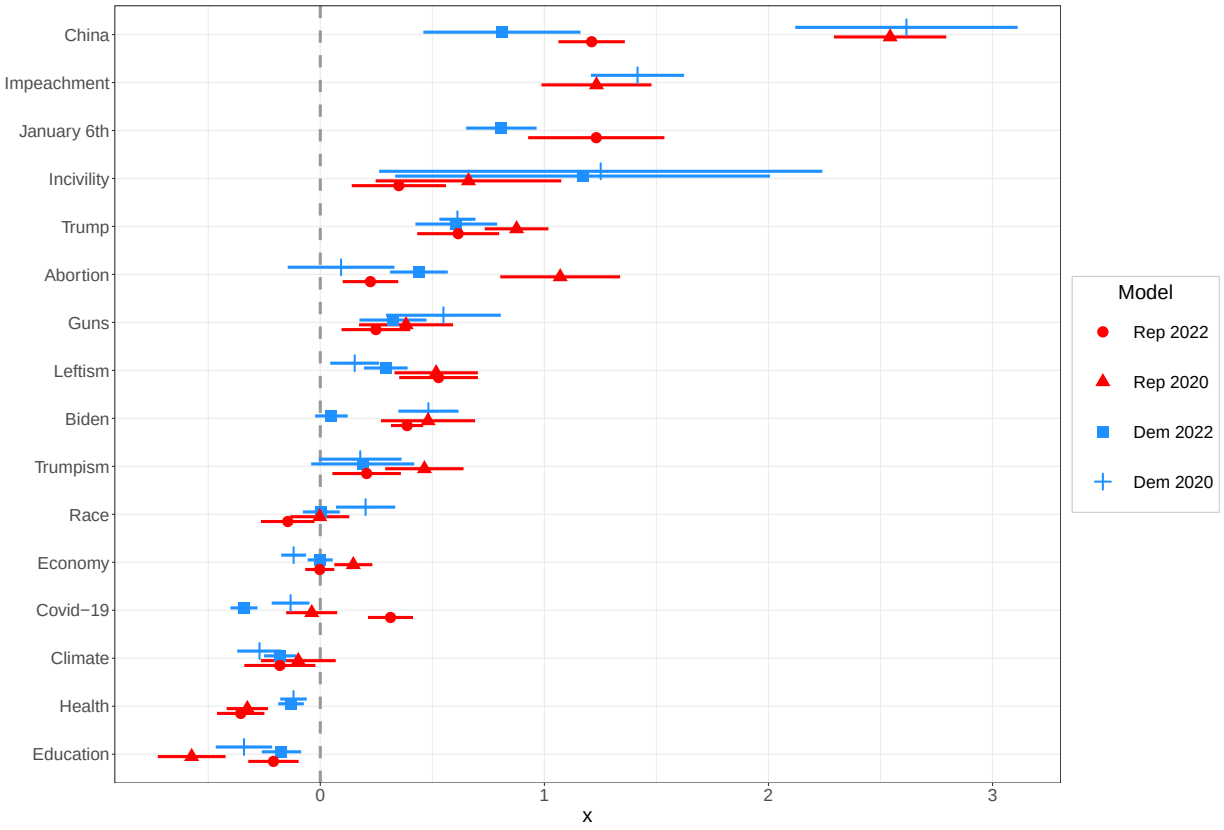


Figure A.19: Association between Topics and Retweets for Incumbents. Coefficients from re-estimating the main negative-binomial models with retweets (rather than total engagement) as the outcome. Whiskers are 95% confidence intervals from candidate-clustered standard errors.

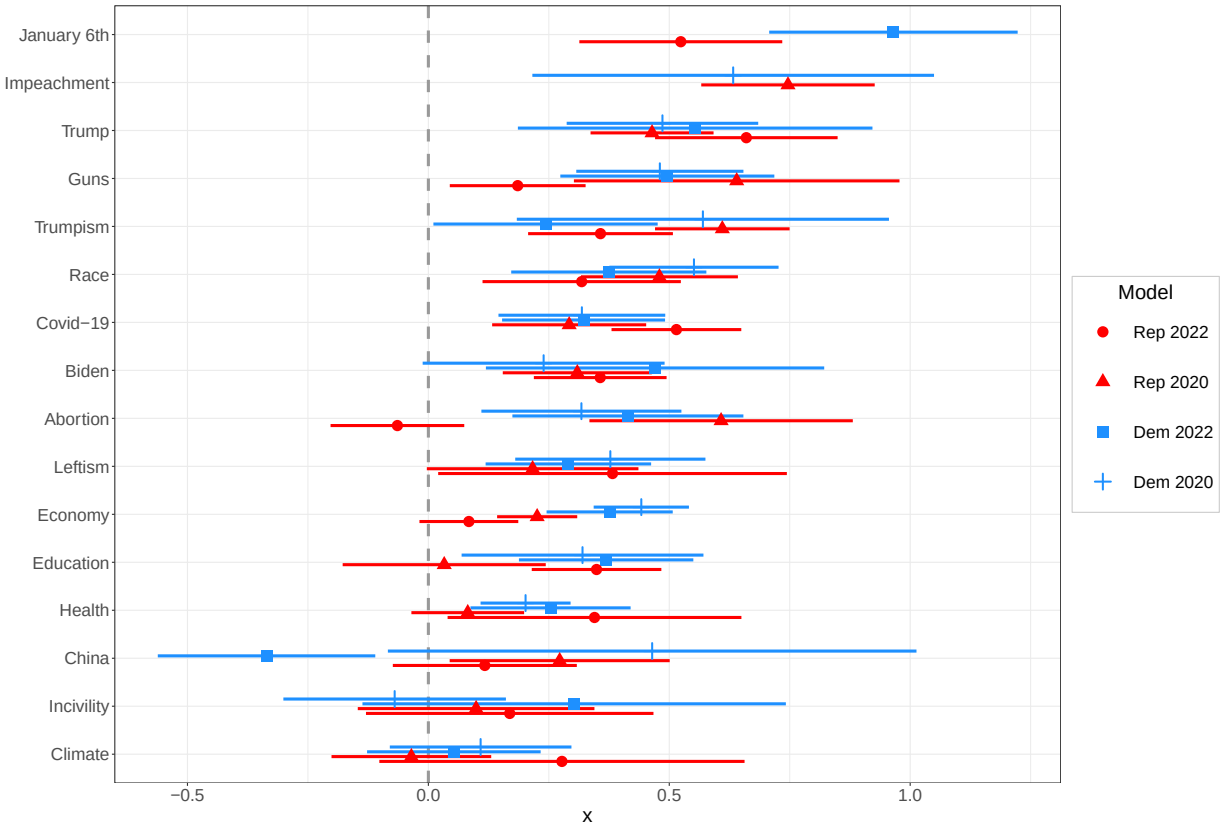


Figure A.20: Association between Topics and Retweets for Challengers. Coefficients from re-estimating the main negative-binomial models with retweets (rather than total engagement) as the outcome. Whiskers are 95% confidence intervals from candidate-clustered standard errors.

H.4 Controlling for Follower Counts

To ensure that changes in candidates' follower counts do not confound our estimates of the rhetorical predictors of engagement, we include logged follower counts as a covariate in each regression.

The Twitter API retrieves follower counts at the time of scraping, rather than when tweets are posted. Because we began scraping in March 2022, we cannot retrieve tweet-level follower counts from the Twitter API for data before that date. We therefore use the Internet Archive's 'Wayback Machine' to recover follower data from January 2019 to February 2022. For low-profile candidates, the Wayback Machine's snapshots are sometimes sparse. We therefore settle for monthly follower counts during this period (01/2019-02/2022). If the Wayback Machine failed to capture a single data snapshot in a month, we used linear interpolation to cover the gap. Thus, for each candidate, we have one follower count data

point for each month.

Even with interpolation, follower data remains missing for many candidate-months. Linear interpolation can only fill months bracketed by observed snapshots on both sides, and the Wayback Machine archives pages roughly in proportion to their traffic, so low-profile candidates often have few or no snapshots at all. Across the 2019–2022 monthly follower files, raw snapshots cover roughly 23–38% of candidate-months, which interpolation raises to 30–51%. Missingness is therefore concentrated among low-profile candidates: relative to the main-text estimation samples, the regressions below exclude roughly 8–12% of incumbents' tweets and 46–50% of challengers' tweets for lack of follower data. For each candidate, we exclude months in which we are missing follower data.

For the 2020 election cycle, this leaves 1,119 candidates and over 1.4 million tweets. Among these, we have 232 Democratic incumbents, 383 Democratic challengers, 188 Republican incumbents, and 316 Republican challengers. For the 2022 election cycle, we have 1,236 candidates and over 898,000 tweets. Among them are 250 Democratic incumbents, 376 Democratic challengers, 217 Republican incumbents, and 393 Republican challengers.

We find that including followers does not change our original topic-engagement estimates: across all topic coefficients, the follower-controlled estimates correlate with the main-text estimates at 0.98, with a maximum absolute deviation of 0.32 on the coefficient (log) scale. This suggests that the key potential time-varying confounder in our main specification (Equation 2), changes in the size of candidates' networks across election cycles, is not a concern.

Table A.4: Association between Topics and Engagement for Incumbents, Controlling for Follower Counts

	Democrats		Republicans	
	2020	2022	2020	2022
Trump	0.52 (0.04)	0.5 (0.11)	0.93 (0.08)	0.59 (0.09)
Guns	0.48 (0.11)	0.25 (0.08)	0.27 (0.1)	0.29 (0.1)
Abortion	0.13 (0.16)	0.4 (0.06)	0.9 (0.18)	0.27 (0.07)
Impeachment	1.58 (0.12)		1.17 (0.13)	
January 6th		0.79 (0.08)		1.25 (0.17)
Race	0.2 (0.08)	-0.07 (0.04)	-0.02 (0.07)	-0.15 (0.06)
Biden	0.5 (0.06)	0.05 (0.04)	0.38 (0.13)	0.26 (0.04)
China	2.02 (0.26)	0.48 (0.17)	1.83 (0.12)	0.77 (0.08)
Covid-19	-0.2 (0.04)	-0.41 (0.03)	-0.11 (0.05)	0.25 (0.05)
Climate	-0.34 (0.05)	-0.23 (0.04)	-0.19 (0.09)	-0.3 (0.08)
Education	-0.34 (0.07)	-0.2 (0.05)	-0.57 (0.08)	-0.21 (0.06)
Economy	-0.19 (0.03)	-0.1 (0.03)	0.06 (0.04)	-0.1 (0.03)
Health	-0.17 (0.03)	-0.2 (0.03)	-0.39 (0.04)	-0.39 (0.05)
Trumpism	0.24 (0.09)	0.22 (0.16)	0.44 (0.08)	0.19 (0.08)
Leftism	0.14 (0.06)	0.27 (0.05)	0.48 (0.11)	0.47 (0.08)
Incivility	1.47 (0.6)	1.22 (0.46)	0.82 (0.23)	0.28 (0.1)
log(Followers)	0.17 (0.07)	-0.03 (0.05)	0.04 (0.04)	0.15 (0.05)
Obs	539483	356527	251401	244097
Pseudo R2	0.11	0.1	0.09	0.09

[1] Clustered standard errors at user-level shown in parentheses.

[2] $(e^{\beta} - 1) * 100$ gives percent change in engagement from mentioning a topic

Table A.5: Association between Topics and Engagement for Challengers, Controlling for Follower Counts

	Democrats		Republicans	
	2020	2022	2020	2022
Trump	0.3 (0.07)	0.46 (0.13)	0.5 (0.06)	0.71 (0.12)
Guns	0.34 (0.11)	0.58 (0.1)	0.37 (0.09)	0.18 (0.09)
Abortion	0.26 (0.11)	0.42 (0.11)	0.24 (0.09)	-0.13 (0.08)
Impeachment	0.44 (0.18)		0.86 (0.16)	
January 6th		0.74 (0.11)		0.41 (0.11)
Race	0.46 (0.11)	0.29 (0.12)	0.56 (0.12)	0.17 (0.09)
Biden	0.34 (0.17)	0.61 (0.19)	0.38 (0.1)	0.32 (0.08)
China	0.03 (0.11)	-0.39 (0.17)	0.15 (0.13)	-0.01 (0.12)
Covid-19	0.21 (0.09)	0.47 (0.12)	0.39 (0.13)	0.6 (0.09)
Climate	-0.07 (0.07)	-0.03 (0.15)	-0.01 (0.15)	0.58 (0.25)
Education	0.32 (0.19)	0.23 (0.09)	0.01 (0.18)	0.36 (0.09)
Economy	0.29 (0.06)	0.39 (0.08)	0.11 (0.05)	0.02 (0.06)
Health	0.09 (0.04)	0.3 (0.11)	-0.06 (0.06)	0.24 (0.16)
Trumpism	0.73 (0.29)	0.2 (0.11)	0.43 (0.1)	0.1 (0.07)
Leftism	0.33 (0.1)	0.25 (0.13)	0.12 (0.15)	0.23 (0.24)
Incivility	0.18 (0.13)	0.29 (0.14)	0.19 (0.1)	0.03 (0.1)
log(Followers)	0.66 (0.07)	0.89 (0.12)	0.44 (0.19)	0.34 (0.1)
Obs	438975	162822	203800	134769
Pseudo R2	0.08	0.08	0.08	0.09

[1] Clustered standard errors at user-level shown in parentheses.

[2] $(e^{\beta} - 1) * 100$ gives percent change in engagement from mentioning a topic